3

Methodology and Pilot Study

This chapter describes the methodological framework underlying the two analytical approaches developed in this thesis. It begins by characterising the data used throughout all experiments: the TCGA-BRCA cohort, comprising matched miRNAs expression profiles and clinical annotations for 272 breast cancer patients (Section 3.1). The subsequent preprocessing pipeline (Section 3.2) transforms this raw data into a clean, modelling-ready feature matrix through a structured sequence of integration, deduplication, missing value handling, encoding, and leakage-aware train/test partitioning steps. The chapter then presents the two core analytical segments. The supervised learning component (Section 3.3) explores the ability of machine learning classifiers to discriminate PAM50 breast cancer subtypes from miRNA expression data, and identifies the minimal set of features most characteristic of each subtype through a two-level validation framework combining XGBoost gain importances and ANOVA F-scores. The unsupervised component (Section 3.4) applies K-Means clustering to assess whether the miRNA expression space contains intrinsic structure that recovers the known subtype partition without any label guidance, and cross-validates the resulting feature signatures against those obtained through the supervised pipeline.

3.1 Data Acquisition and Characterisation

3.1.1 Data Source

The dataset used in this work originates from **The Cancer Genome Atlas Breast Invasive Carcinoma** (TCGA-BRCA) project [61], a large-scale, publicly available multi-omics repository maintained by the National Cancer Institute. TCGA-BRCA represents one of the most comprehensively characterized breast cancer cohorts in the biomedical literature, providing standardised genomic, transcriptomic, and clinical data for a large cohort of patients profiled with uniform experimental protocols. Its public availability, scale, and annotation quality makes it the canonical benchmark dataset for computational breast cancer subtyping studies, and it has been used as the primary data source in the majority of related work discussed in Chapter 2.

Two complementary data releases were obtained from the TCGA-BRCA portal and used jointly throughout this work: a **microRNA expression matrix**, containing quantified expression levels for 888 miRNA features across 272 patient samples, and a **clinical data table**, containing 37 variables per patient across a total of 463 records. These two sources are described in detail in Sections 3.1.3 and 3.1.4 respectively. Their integration into a unified modelling dataset is described as part of the preprocessing pipeline in Section 3.2.

3.1.2 The Working Cohort

The full clinical release and the miRNA expression data contain records for 463 and 272 patients, respectively. However, not all patients in the clinical table have a corresponding miRNA expression profile available in the molecular data release, which will drastically decrease our working cohort given that our objective is to find meaningful associations between miRNA, (potentionally) clinical features and subtypes of BC. After restricting the cohort to individuals for whom matched miRNA expression profiles existed, a **working cohort of 272 samples** was obtained with the intersection of the two data sources. All 272 samples correspond to primary solid tumour tissue (`SAMPLE_TYPE = Primary`) from mostly female patients (with over 99% representation) , consistent with the epidemiology of breast cancer as a predominantly female disease.

Table 3.1: Detailed Descriptive Statistics of the TCGA Dataset

Statistic	**Value**
General Metrics	
Total miRNA Samples (N)	272
Total Clinical Samples	463
Number of microRNAs	888
Clinical Attributes	37
Clinical Data Insights	
Age Range	26–90 years

The classification target throughout this work is the **PAM50 intrinsic subtype**, a well-established gene-expression-based taxonomy that partitions breast cancer into five biologically distinct classes: Luminal A, Luminal B, HER2-enriched, Basal-like and Normal-like (we have already explored the molecular and biological characteristics of each subtype in Section 2.2.1). The subtype distribution within the working cohort is shown in Table 3.2 and the accompanying Figure 3.1.

The class distribution reveals a pronounced imbalance that is characteristic of breast cancer populations in clinical and research settings. Luminal A is the dominant subtype, comprising 117 of 272 patients (43.0%), while Normal-like is severely underrepresented with only 5 samples (1.8%). This asymmetry has direct consequences for model training and evaluation: a naive classifier that always predicts Luminal A would achieve over 43%

PAM50 Subtype	**Count**	**%**
Luminal A	117	43.0
Luminal B	75	27.6
Basal-like	45	16.5
HER2-enriched	30	11.0
Normal-like	5	1.8
Total	**272**	**100.0**

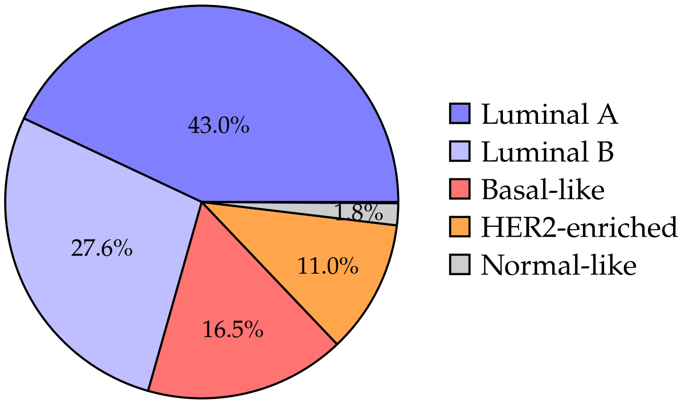


Figure 3.1: PAM50 subtype distribution across the 272-sample working cohort. Luminal A is the dominant class (43%), while Normal-like is severely underrepresented ($n = 5$; 1.8%). This pronounced class imbalance is a central challenge addressed in the modelling strategy in Section 3.2.6.

accuracy without learning anything about the underlying biology. The modelling decisions described in Section 3.3 are informed by this imbalance, and stratified sampling is used throughout to ensure that each experimental fold preserves the original class proportions.

3.1.3 Clinical Data

The clinical data table contains 37 variables per patient. Several of these are administrative or redundant fields (e.g., `sampleId`, `patientId`, `studyId`, `CANCER_TYPE`), that carry no predictive information for subtype classification and were excluded during preprocessing. A (pre) feature selection process was applied to the clinical data, resulting in a subset of 20 variables. These were retained based on their perceived relevance and informative value for the predictive model. These features are listed in Table 3.3, along with their data type, possible values, and a brief description of what each variable represents clinically.

The 20 retained features span a broad range of clinical and molecular dimensions. The three receptor statuses, `ER_STATUS`, `PR_STATUS`, and `HER2_STATUS`, are the canonical markers used in clinical breast cancer diagnosis and are intrinsically aligned with the PAM50 subtype labels: for example, the Basal-like subtype is largely composed of triple-negative tumours ($\mathrm{ER}^-/\mathrm{PR}^-/\mathrm{HER2}^-$), a well-documented biological correspondence that provides a natural consistency check for classifier outputs. The TNM staging variables (`TUMOR_STAGE`, `NODES`, `METASTASIS`) encode the anatomical extent of the disease at diagnosis, while the genomic variables (`MUTATION_COUNT`, `TMB_NONSYNONYMOUS`, `FRACTION_GENOME_ALTERED`) reflect tumour mutational characteristics that may (or may not) correlate with subtype-specific oncogenic programmes. The integrative cluster assignments (`CN_CLUSTER`, `METHYLATION_CLUSTER`, `MIRNA_CLUSTER`) are derived from TCGA's own multi-omics analyses and encode cross-modal molecular patterns that may complement the raw miRNA expression signal.

Data Descriptive Statistics A descriptive overview of selected clinical characteristics within the working cohort is as follows. Patient ages range from 26 to 90 years, with a mean of 59.3 ± 13.3 years (median = 60), consistent with the known epidemiology of breast cancer as predominantly a post-menopausal disease. Oestrogen receptor positivity was recorded in 77.2% of patients ($n = 210$), progesterone receptor positivity in 64.7% ($n = 176$), and HER2 amplification in 13.6% ($n = 37$). These proportions are broadly representative of an unselected breast cancer population. Tumour stage at diagnosis was dominated by T2 lesions (61.4%, $n = 167$), followed by T1 (19.5%), T3 (12.9%), and T4 (5.1%). From a survival perspective, 89.0% of patients ($n = 242$) were alive at the time of last follow-up, with 30 recorded deaths.

Several clinical features contain missing values that require attention during preprocessing. The integrative cluster columns are missing for a large fraction of patients (approximately 76 and 73 records respectively) and were therefore excluded from the modelling feature set prior to the cohort restriction step. These features were:

- `INTEGRATED_CLUSTERS_NO_EXP`;
- `INTEGRATED_CLUSTERS_UNSUP_EXP`;
- `INTEGRATED_CLUSTERS_WITH_PAM50`;
- `RPPA_CLUSTER`.

Among the 20 retained features, missing values are sparse (at most 4 records for `METASTASIS` and `METASTASIS_CODED`). The strategy for handling these residual missing values is described in Section 3.2.

3.1.4 MicroRNA Expression Data

The miRNA expression matrix contains quantified expression levels for **888 miRNA features** measured across the 272 patient samples in the working cohort. Each entry x_{ij} represents the expression level of miRNA j in patient i, where higher values indicate greater expression abundance in the tumour tissue. This constitutes a canonical **high-dimensional, low-sample-size (HDLSS)** setting: the number of features ($p = 888$) greatly exceeds the number of observations ($n = 272$), a regime that poses well-known challenges for statistical learning and motivates the feature selection strategies explored throughout this work.

The miRNA Redundancy Problem

A critical characteristic of the raw miRNA expression data, which becomes apparent upon close examination, is the presence of widespread feature redundancy. The 888 features in the raw matrix do not represent 888 biologically distinct microRNAs. Instead, a large number of entries share the same *base identifier* - for example, `MIR-758/758`, `MIR-758/3P`,

Table 3.3: Clinical features retained for modelling, with data type, possible values, and clinical meaning.

Feature	Type	Values	Description
AGE	Integer	26–90	Patient age at diagnosis (years).
ER_STATUS	Categorical	Positive, Negative, Indeterminate, Not Available	Oestrogen receptor status assessed by immunohistochemistry.
PR_STATUS	Categorical	Positive, Negative, Indeterminate, Not Available	Progesterone receptor status assessed by immunohistochemistry.
HER2_STATUS	Categorical	Positive, Negative, Equivocal	HER2 receptor amplification status.
TUMOR_STAGE	Ordinal	T1, T2, T3, T4, TX	Primary tumour size/extent per TNM staging.
TUMOR_T1_CODED	Binary	T1, T_Other	Binarised tumour stage: T1 vs. all others.
NODES	Ordinal	N0, N1, N2, N3	Regional lymph node involvement per TNM staging.
NODE_CODED	Binary	Positive, Negative	Binarised lymph node involvement.
METASTASIS	Binary	M0, M1	Presence of distant metastasis per TNM staging.
METASTASIS_CODED	Binary	Positive, Negative	Binarised metastasis indicator.
CONVERTED_STAGE	Categorical	No_Conversion, Stage I, Stage IIA, Stage IIB, Stage IIIA, Stage IIIB, Stage IIIC, Stage IV	Overall AJCC composite stage derived from TNM.
FRACTION_GENOME _ALTERED	Continuous	$[0, 1]$	Proportion of the genome showing copy-number alterations.
MUTATION_COUNT	Integer	≥ 0	Total number of somatic mutations detected.
TMB_NONSYNONYMOUS	Continuous	≥ 0	Tumour mutational burden (non-synonymous mutations per Mb).
OS_MONTHS	Continuous	≥ 0	Overall survival time from diagnosis to last follow-up (months).
OS_STATUS	Binary	0:LIVING, 1:DECEASED	Patient vital status at last follow-up.
CN_CLUSTER	Integer (cluster)	1–5	Copy-number cluster assignment from TCGA integrative analysis.
METHYLATION_CLUSTER	Integer (cluster)	1–5	DNA methylation cluster from TCGA integrative analysis.
MIRNA_CLUSTER	Integer (cluster)	1–7	miRNA expression cluster from TCGA integrative analysis.
SIGCLUST_UNSUPERVISED _MRNA	Integer	Signed integer scores	Significance of cluster assignment from unsupervised mRNA analysis.

and `MIR-758/5P` all derive from the same precursor microRNA (MIR-758), differing only in the strand arm or isoform designation. Across the full matrix, 888 Hugo gene symbols map to only 442 unique base identifiers, meaning that 446 features are variants of a base miRNA already present in the data.

More importantly, upon inspection of the actual expression values, many of these variants are found to carry **identical expression profiles** across all 272 patients - that is, for a given patient, `MIR-758/758`, `MIR-758/3P`, and `MIR-758/5P` all report exactly the same numerical value. This redundancy is attributable to a fundamental limitation of the sequencing and quantification technology used to generate the TCGA-BRCA expression data: the platform lacks the resolution to distinguish between closely related isoforms of the same precursor miRNA, and consequently assigns the same measured abundance to all of them. From a machine learning perspective, these duplicated features add no discriminative information whatsoever (they are mathematically identical columns in the feature matrix) and including them in the model would artificially inflate the apparent feature space while introducing collinearity that could distort feature importance estimates.

This observation directly motivates one of the first and most consequential preprocessing decisions in the pipeline: the identification and removal of duplicate expression profiles, described in detail in Section 3.2. As reported in our published study I, this deduplication step reduces the miRNA feature count from 888 to 442, eliminating 446 non-informative redundant features while retaining the full unique expression signal. The resulting feature matrix is both more compact and more interpretable, with each remaining miRNA representing a genuinely distinct molecular measurement.

Now that we have characterized the data sources and the working cohort, we can proceed to the pre-processing pipeline that transforms the raw data into a clean, numerical feature matrix suitable for machine learning. The next section describes this pipeline in detail, including the handling of missing values, feature selection decisions, and the critical steps taken to avoid data leakage between training and test partitions.

3.2 Data Pre-processing Pipeline

Before any learning algorithm was applied, the raw data sources described in Section 3.1 underwent a structured sequence of transformations designed to produce a clean, fully numerical, and modelling-ready feature matrix. The pipeline was designed with two overriding principles in mind: preserving the biological authenticity of the data at every step, and strictly preventing any statistical information from the test set from influencing the training process (**Data Leakage Prevention**). The full procedure is summarised schematically in Figure 3.2, and each step is described in the subsections below.

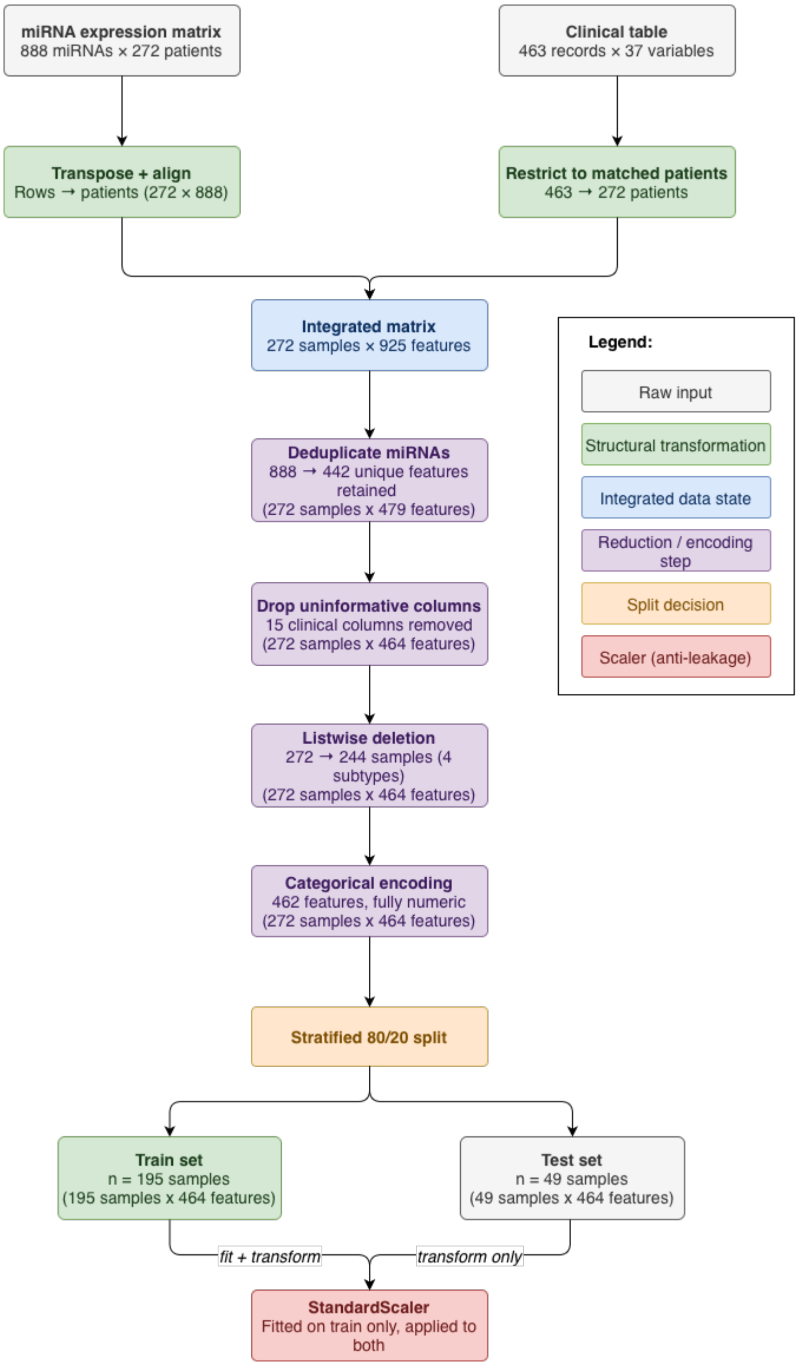


Figure 3.2: Schematic overview of the data pre-processing pipeline. The raw miRNA expression matrix and clinical data table are transformed through a series of steps including integration, deduplication, feature selection, and encoding, culminating in a stratified train-test split and final scaling. Each step is designed to preserve data integrity while preparing the dataset for robust machine learning modelling.

3.2.1 Data Integration

The two raw data sources arrived in structurally incompatible orientations. The miRNA expression matrix followed the conventional bioinformatics layout: features (miRNAs) as rows and patients as columns, yielding a matrix of 888×275 (where 3 of the 275 columns are metadata fields: `entrezGeneId`, `hugoGeneSymbol`, and `type`). The clinical data table, by contrast, followed the standard tabular layout with patients as rows and variables as columns (463×37). Before any join could be performed, the miRNA matrix had to be transformed to match this convention. The transformation proceeded in four steps:

(i) the metadata columns `entrezGeneId` and `type` were dropped, retaining only the gene symbol and expression values;

(ii) the matrix was transposed, placing patients on the rows and miRNAs on the columns;

(iii) the first row of the transposed matrix (which contained the miRNA identifiers) was promoted to serve as the column header;

(iv) the row index was renamed to `sampleId` to establish a common join key with the clinical table.

The two tables were then merged via an **inner join** on `sampleId`. The inner join naturally restricts the combined dataset to patients for whom both miRNA expression data and clinical annotations are available simultaneously, yielding an integrated matrix of **272 samples** and $888 + 37 = 925$ features prior to any reduction step. The 191 clinical records without matched miRNA profiles were discarded at this stage.

3.2.2 miRNA Deduplication

As described in Section 3.1.4, the 888 miRNA features in the raw matrix do not represent 888 biologically independent measurements. A substantial fraction of entries share the same precursor base identifier, for example, `MIR-758/758`, `MIR-758/3P`, and `MIR-758/5P` (visually explained in Table 3.4). These carry **identical numerical expression values** across all 272 samples. This redundancy comes from the limited sensibility of the extraction technique used to collect these miRNA expression values which cannot distinguish between closely related isoforms of the same precursor, and consequently assigns the same measured abundance to all of them.

From a machine learning standpoint, retaining these duplicate columns is not merely wasteful: **it is actively harmful**. Identical features contribute zero additional information to the model while artificially inflating the apparent dimensionality of the feature space, exacerbating the curse of dimensionality in an already data-sparse regime. They also introduce perfect multicollinearity, which can distort coefficient estimates in linear models and destabilise feature importance rankings in tree-based models.

To address this, all miRNA features sharing identical expression profiles across all samples were identified and all but one representative were removed. The deduplication was applied by grouping features according to their base identifier and retaining only those with a unique expression vector. This step reduced the miRNA feature count from

Table 3.4: Example of miRNA identifier variants sharing identical expression values across patients. Each row corresponds to a distinct annotation of the same base identifier (MIR-744), yet all variants carry the exact same expression value for every patient - here illustrated for two example patient IDs (TCGA-A1-A0SE-01 and TCGA-A1-A0SH-01). This redundancy across variant annotations motivates the filtering step applied prior to modelling.

Hugo Gene Symbol	Type	TCGA-A1-A0SE-01	TCGA-A1-A0SH-01
MIR-744/744*	miRNA	3.67519164	3.03321064
MIR-744/744	miRNA	3.67519164	3.03321064
MIR-744/3P	miRNA	3.67519164	3.03321064
MIR-744/5P	miRNA	3.67519164	3.03321064

888 to 442, eliminating 446 redundant features without discarding any unique biological signal.

3.2.3 Removal of Uninformative Clinical Features

When we look at the 37 original clinical variables, 15 were removed prior to modelling on the grounds summarised in Table 3.5. These columns fell into four categories:

- redundant patient identifiers that carry no predictive signal (patientId);
- zero-variance fields that take a single value across all samples and therefore provide no discriminative information (studyId, CANCER_TYPE, CANCER_TYPE_DETAILED, ONCOTREE_CODE, SAMPLE_TYPE, SOMATIC_STATUS, SAMPLE_COUNT, SEX);
- high-missingness fields where the proportion of missing values rendered the feature unreliable (RPPA_CLUSTER at ≈24% missing;
- INTEGRATED_CLUSTERS_NO_EXP, INTEGRATED_CLUSTERS_UNSUP_EXP, and INTEGRATED_CLUSTERS_WITH_PAM50 each at ≈25% missing);
- task-irrelevant derived features (SIGCLUST_INTRINSIC_MRNA, a cluster assignment derived from mRNA rather than miRNA data, and SURVIVAL_DATA_FORM, a near-zero-variance administrative field).

After these removals, 20 clinical features were retained alongside the 442 deduplicated miRNA expression features, for an intermediate feature matrix of 272 samples × 462 features (plus the sampleId identifier and the PAM50_SUBTYPE target).

3.2.4 Handling Missing Values & Low Representation Classes

Following the elimination of high-missingness columns, a small residual number of samples still contained at least one missing value across the remaining clinical features. Specifically, the affected fields and their missing-value counts were: FRACTION_GENOME_ALTERED

Table 3.5: Clinical columns removed during pre-processing, with the rationale for each exclusion.

Column	Reason	Detail
patientId	Redundant identifier	Duplicate of sampleId.
studyId, CANCER_TYPE, CANCER_TYPE_DETAILED, ONCOTREE_CODE, SAMPLE_TYPE, SOMATIC_STATUS, SAMPLE_COUNT, SEX	Zero variance	Single value across all 272 samples (all primary female BRCA).
RPPA_CLUSTER	High missingness	≈24% missing values.
INTEGRATED_CLUSTERS_NO_EXP, INTEGRATED_CLUSTERS_UNSUP_EXP, INTEGRATED_CLUSTERS_WITH_PAM50	High missingness	≈25% missing values each.
SIGCLUST_INTRINSIC_MRNA	Task-irrelevant	Derived from mRNA, not miRNA; circular with target.
SURVIVAL_DATA_FORM	Near-zero variance	Administrative field with negligible variation.

($n = 2$), HER2_STATUS ($n = 1$), METASTASIS_CODED ($n = 4$), TUMOR_T1_CODED ($n = 3$), and METASTASIS ($n = 4$). The miRNA expression matrix itself contained no missing values.

Deciding how to handle missing data is never a purely technical choice: the appropriate strategy depends on the nature of the data, the mechanism generating the missingness, and the downstream use. In computational studies applied to purely numerical or synthetic data, some imputation methods such as mean substitution, k-nearest neighbour imputation, or generative approaches like MICE [13] are standard and often effective. In a biological dataset such as this one, however, the situation is qualitatively different. Clinical measurements such as metastasis status, HER2 assessment, or genomic alteration fraction reflect real, heterogeneous biological processes in individual patients. Filling in a missing HER2 status with the cohort mean, or inferring a missing metastasis record from neighbouring samples, risks fabricating a clinical picture that does not correspond to any real biological state. Imputed values could mislead the model into learning artificial patterns (in effect, training on synthetic patients that never existed). Given the already limited size of the cohort, introducing even a small number of biologically implausible observations could meaningfully distort the learned decision boundaries.

Given that the number of affected samples is small relative to the overall cohort, and given the potential for imputed values to introduce bias, the affected samples were removed via **listwise deletion**. The miRNA expression matrix contained no missing values, so this step exclusively affected rows with incomplete clinical annotations. In total, 28 samples were removed, **reducing the cohort from 272 to 251 samples.** This represents a loss of 10.3% of the working cohort, a non-trivial reduction, but one that is justified by the imperative of data integrity in a biomedical context. In addition, at the encoding stage described below, the 5 remaining Normal-like samples were excluded from the supervised experiments due to their insufficient representation for reliable learning and evaluation, bringing the final modelling cohort to **244 samples** across four PAM50 subtypes: Luminal A ($n = 103$), Luminal B ($n = 71$), Basal-like ($n = 41$), and HER2-enriched ($n = 29$).

3.2.5 Categorical Encoding

Machine learning algorithms operate on numerical representations: they compute distances, gradients, and split criteria over real-valued feature spaces. Categorical variables, those that take a finite set of discrete named values rather than continuous numbers, cannot be directly ingested by most learning algorithms without first being converted to a numerical form. Among the 20 retained clinical features, 11 are categorical and required explicit encoding, while the remaining 9 are already numerical and required no transformation: `CN_CLUSTER`, `FRACTION_GENOME_ALTERED`, `METHYLATION_CLUSTER`, `MIRNA_CLUSTER`, `MUTATION_COUNT`, `SIGCLUST_UNSUPERVISED_MRNA`, `TMB_NONSYNONYMOUS`, `AGE`, and `OS_MONTHS`.

Two fundamentally different types of categorical variables appear in this dataset, and they call for different encoding strategies:

- **Ordinal variables** carry a natural rank ordering among their categories: the categories are not merely distinct names but lie on a meaningful scale. For these variables, integer codes were assigned to *preserve the natural clinical ordering*, so that the numerical distance between codes reflects the conceptual distance between categories. For example, `NODES` (lymph node involvement) follows the TNM classification N0 < N1 < N2 < N3 in order of increasing severity, and was encoded accordingly as $\{0, 1, 2, 3\}$. Similarly, `TUMOR_STAGE` was encoded as T1→1, T2→2, T3→3, T4→4 to preserve the size-based ordering.

- **Nominal variables** have categories that are qualitatively distinct but carry no inherent ordering. For binary nominal variables (such as `METASTASIS_CODED` (Negative/Positive), `NODE_CODED`, `METASTASIS` (M0/M1), and `OS_STATUS` (0:LIVING/1:DECEASED)) standard 0/1 integer encoding was applied, since with only two categories the choice of which value maps to 0 or 1 is arbitrary and carries no structural implication. For multi-valued nominal variables with no meaningful ordering, a judgment call was required. `ER_STATUS` and `PR_STATUS`, for instance, each take four values (Negative, Positive, Performed but Not Available, Indeterminate), and `HER2_STATUS` takes three (Negative, Equivocal, Positive). These were encoded with manual integer mappings. It should be noted that for `HER2_STATUS`, there is a partial clinical ordering (Negative < Equivocal < Positive), and the encoding 0/1/2 was chosen to reflect this.

The complete encoding scheme for all 11 categorical features is listed in Table 3.6. We can see that the target variable PAM50_SUBTYPE was encoded at this stage with the following mapping: Basal-like→0, HER2-enriched→1, Luminal A→2, Luminal B→3. Normal-like samples received no mapping and were thus excluded from the supervised modelling cohort, as noted in Section 3.2.4. Upon completion of the encoding step, the dataset had the shape of **244 samples** described by **462 features** distributed in 20 clinical variables and 442 miRNA expression values - all represented as real numbers now.

Table 3.6: Encoding scheme for the 11 categorical clinical features. Ordinal features are highlighted; the ordering of the integer codes reflects the natural clinical ranking.

Feature	**Type**	**Encoding**
NODES	Ordinal	N0→0, N1→1, N2→2, N3→3
TUMOR_STAGE	Ordinal	T1→1, T2→2, T3→3, T4→4
CONVERTED_STAGE	Ordinal	No_Conversion→0, Stage I→2, Stage IIA→1, Stage IIB→3, Stage IIIA→5, Stage IIIB→7, Stage IIIC→4, Stage IV→6
HER2_STATUS	Ordinal*	Negative→0, Equivocal→1, Positive→2
ER_STATUS	Nominal	Negative→0, Performed but Not Available→1, Positive→2, Indeterminate→3
PR_STATUS	Nominal	Negative→0, Positive→1, Performed but Not Available→2, Indeterminate→3
TUMOR_T1_CODED	Binary	T_Other→0, T1→1
NODE_CODED	Binary	Negative→0, Positive→1
METASTASIS_CODED	Binary	Negative→0, Positive→1
METASTASIS	Binary	M0→0, M1→1
OS_STATUS	Binary	0:LIVING→0, 1:DECEASED→1

* Partial ordering exists clinically; encoding reflects severity progression.

3.2.6 Train/Test Split and Prevention of Data Leakage

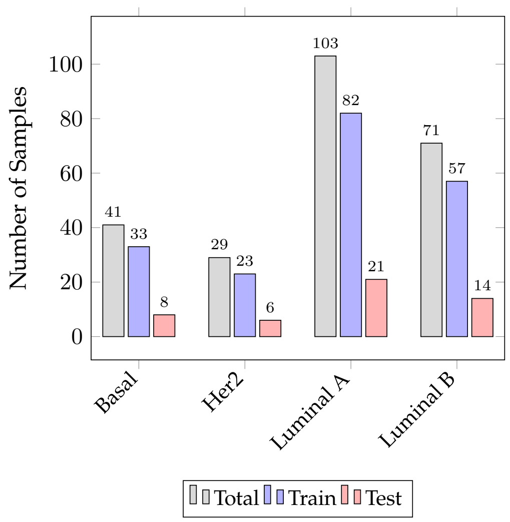
The final processing step before model training was the partition of the dataset into a training set and a held-out test set, followed by feature standardisation. The order in which these operations are performed is not a minor implementation detail: it is a methodologically critical design decision that determines whether reported performance metrics are trustworthy estimates of generalisation ability or inflated artefacts of inadvertent information leakage.

Stratified Splitting & Data Leakage Prevention

To ensure a robust evaluation, the dataset was divided into an **80% training set** (195 samples) and a **20% test set** (49 samples) using stratified random sampling. Stratification was essential to maintain the proportion of each PAM50 subtype across both partitions, thereby preventing the accidental over-representation or under-representation of minority classes: a critical safeguard given the class imbalance documented in Section 3.1.2. The resulting class distributions, which demonstrate the consistency of this split, are shown in Figure 3.3b.

Property	**Value**
Dimensionality	
Total samples	244
miRNA features	442
Clinical features	20
Total features	462
Feature-to-sample ratio (p/n)	1.89
Data quality	
Missing values	0
miRNAs with variance > 1.0	250 (56.6%)
miRNA expression ($\log_2$ scale)	
Range	$[-2.37, 18.66]$
Mean ± std	4.87 ± 3.99
Median	4.07
Values < 0 (low-abundance artefact)	7.1%
Clinical continuous features	
Age (years)	59.7 ± 13.4 [26, 89]
Mutation count	45.7 ± 36.3 [10, 223]
Overall survival (months)	34.3 ± 41.4 [0, 234]
Fraction genome altered	0.34 ± 0.21 [0.00, 0.95]
TMB non-synonymous	1.53 ± 1.21 [0.30, 7.40]

(a) Summary statistics of the final dataset.



(b) Subtype distribution histogram.

Figure 3.3: Overview of the final modelling dataset: (a) detailed summary of clinical and molecular features; (b) distribution of PAM50 subtypes across the full dataset and the stratified train/test partitions.

This partitioning serves as the primary defense against **data leakage**, which occurs when information from outside the training set is used during the learning process. Leakage is a frequent source of overly optimistic performance estimates, as the model inadvertently learns patterns from data that would be unavailable at inference time. In this pipeline, the prevention of leakage was strictly enforced, particularly regarding **scaling leakage**. This common pitfall manifests if a *StandardScaler* is fitted on the full dataset before the split; in such cases, the scaling parameters (the mean μ and standard deviation σ per feature) would be derived from both training and test samples. By ensuring the split occurs before any statistic-based transformation, the test set remains truly unseen, preserving the independence assumption required for a meaningful evaluation.

To prevent this, the following protocol was strictly enforced throughout all experiments:

1. The 80/20 stratified split was performed **first**, before any scaling parameter was computed.

2. A *StandardScaler* was then fitted **exclusively on the training fold**, computing the per-feature mean μ_{train} and standard deviation σ_{train} from training samples only.

3. The transformation derived from the training fold was then applied **without re-fitting** to both the training and test folds:

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}} \tag{3.1}$$

   where z is the standardised feature value and x is the original value. The test fold is thus transformed using training statistics, simulating the situation at deployment time where the model encounters data whose distribution it has not observed.

Standardisation was applied to two groups of features: the five continuous clinical variables with heterogeneous scales (`MUTATION_COUNT`, `SIGCLUST_UNSUPERVISED_MRNA`, `TMB_NONSYNONYMOUS`, `AGE`, `OS_MONTHS`), and all 442 miRNA expression columns. The remaining clinical features, those already encoded as small integers (representing ordinal or binary categories) were left unscaled as their values are already bounded and comparable in magnitude.

Raw miRNA expression values span the range $[-2.37, 18.66]$ on the $\log_2$ scale, with a mean of 4.97 and a standard deviation of 4.03. Clinical continuous features operate on entirely different numerical scales: `MUTATION_COUNT` ranges from 10 to 223, `OS_MONTHS` from 0 to 234, and `FRACTION_GENOME_ALTERED` from 0 to 0.95. Without standardisation, algorithms sensitive to feature magnitude, such as distance-based methods and gradient-based optimisers, would disproportionately weight high-magnitude features irrespective of their actual predictive value.

3.3 Discriminative Subtyping

As described in Section 3.2, the dataset used throughout this work consists of **244 patient samples** sourced from TCGA [61], each characterised by two distinct feature modalities: **445 miRNA expression features** and **17 clinical features** (including hormonal receptor status, copy-number cluster assignments, and methylation cluster labels), together with a PAM50-derived subtype label (*Basal-like*, *HER2-enriched*, *Luminal A*, or *Luminal B*). After the preprocessing pipeline detailed in the previous chapter, the data was organised into training and test partitions that are used consistently across all experiments reported here.

The present chapter focuses on the **supervised learning component** of this thesis. The central objective is twofold: first, to determine whether machine learning classifiers can reliably distinguish breast cancer subtypes from miRNA expression data alone, and more importantly, whether the addition of clinical covariates improves that discrimination; second, and most crucially, to **identify the minimal set of miRNA (and clinical) features that best (minimally) discriminate each subtype**. This second objective aligns directly with the broader biological question motivating this work: **which miRNAs carry**

subtype-discriminative information, and can machine learning provide a computationally grounded way to answer it?

3.3.1 Initial Baselines Assessment

An initial exploratory phase was conducted with three classifiers - **Decision Tree**, **XGBoost**, and **DIABLO** - to characterise the difficulty of the multiclass PAM50 subtype prediction task and identify the most promising architecture for further development. These three models were selected to probe different ends of the complexity spectrum: a single-split decision surface, a full gradient-boosted ensemble, and a sparse multi-omics latent-variable model designed specifically for heterogeneous data integration [57]. All three were evaluated in their default or near-default configurations, without hyperparameter tuning, using both feature configurations (miRNA only and miRNA + clinical data).

A criterion that shaped the selection equally alongside predictive performance was **interpretability**. The ultimate aim of this supervised approach is not classification per se, but the identification of the miRNA features that best characterise each subtype. A model that classifies accurately but obscures its decision process provides limited biological insight. This requirement favoured models with accessible and meaningful feature importance mechanisms, a point that would later prove decisive in choosing XGBoost as the final architecture.

3.3.2 Baseline Results

Table 3.7 reports the per-class classification performance of the three baseline models on the held-out test set under the miRNA + clinical configuration. Across all models, Basal-like was the most consistently well-classified subtype, while HER2-enriched and Luminal B emerged as the most challenging - a pattern that would persist through subsequent refinements.

Table 3.7: Baseline model performance on the held-out test set (miRNA + Clinical Data). Classes: 0 = Basal-like, 1 = HER2-enriched, 2 = Luminal A, 3 = Luminal B.

Model	Metric	Basal	HER2	Lum A	Lum B	Macro Avg	Accuracy
Decision Tree	Precision	0.80	0.60	0.62	0.46	0.62	0.61
	Recall	1.00	**0.50**	0.62	0.43	0.64	
	F1-score	0.89	**0.55**	0.62	0.44	0.62	
DIABLO	Precision	0.89	0.25	0.68	0.45	0.57	0.63
	Recall	**1.00**	0.17	0.81	0.36	0.58	
	F1-score	0.94	0.20	0.74	0.40	0.57	
XGBoost	Precision	**1.00**	**1.00**	**0.71**	**0.73**	**0.86**	**0.78**
	Recall	**1.00**	0.33	**0.95**	**0.57**	**0.71**	
	F1-score	**1.00**	0.50	**0.82**	**0.64**	**0.74**	

XGBoost already achieved the strongest baseline performance, suggesting that an ensemble approach was well-suited to this high-dimensional setting. DIABLO, despite being purpose-built for multi-omics integration, was limited by the relatively small cohort size ($n = 244$), which constrains the reliable estimation of its cross-block covariance structure. These observations established XGBoost as the primary candidate for systematic refinement in the next phase due to its classification performance and interpretability.

3.3.3 Comparative Evaluation with Hyperparameter Optimisation

Building on the baseline assessment, a more systematic comparative evaluation was conducted. **Logistic Regression** was introduced at this stage as a regularised linear baseline - a standard reference point in classification benchmarks whose presence anchors the comparison by establishing the performance ceiling of a linear decision boundary on this dataset. Together with the refined versions of Decision Tree, DIABLO, and XGBoost, the four models were each subjected to stratified cross-validated grid search, with macro F1-score as the optimisation criterion.

For XGBoost, the parameters searched were:

- **Number of estimators** (`n_estimators`): number of boosting rounds (trees) in the ensemble. A larger value allows a more expressive model at the cost of increased training time and potential overfitting.
- **Learning rate** (`learning_rate`): the shrinkage applied to each tree's contribution. Smaller values require more trees but generally improve generalisation.
- **Maximum tree depth** (`max_depth`): controls model complexity. Deeper trees capture higher-order interactions but overfit more easily.
- **Row subsampling** (`subsample`): the fraction of training samples used per boosting round, introducing stochasticity to reduce variance.
- **Column subsampling** (`colsample_bytree`): the fraction of features randomly selected at each boosting round, analogous to the feature bagging in Random Forests.

After computing the best configuration for our model paired with stratified cross-validation with macro F1-score as the optimisation criterion, we've ended up with the following configuration:

Hyperparameter	**Value**
`n_estimators`	50
`learning_rate`	0.3
`max_depth`	10
`subsample`	0.85
`colsample_bytree`	1.0
`eval_metric`	`mlogloss`
`random_state`	42

The results achieved using this configuration will be analysed in detail in Section 3.3.6, with a complementary exploration available in Section 3.3.5 where we evaluate the statistical significance of the features selected by XGBoost.

3.3.4 Fine tunned results

Now that we've fine tunned our models, with special focus on XGBoost, we can evaluate its performance on the same test set as before, but now with models that have been fine tunned. Table 3.8 reports the results of all four models after hyperparameter optimisation under the miRNA + clinical configuration - given that this configuration is the one that we've found to be the most performant for the heterogeneous dataset.

Table 3.8: Heterogeneous models (miRNAs + Clinical Data). Results are shown in percentage.

	Overall				**Basal**	**HER-2**	**Lum A**	**Lum B**
Model	P	R	F1	Acc	F1			
DIABLO	78.47	72.62	73.82	79.59	**100.00**	40.00	85.71	69.57
Decision Tree	62.01	63.69	62.45	61.22	88.89	54.55	61.90	44.44
Logistic Regression	77.22	75.60	75.57	79.59	94.12	54.55	**86.96**	66.67
XGBoost	**90.06**	**79.17**	**82.17**	**83.67**	**100.00**	**66.67**	85.11	**76.92**

The results confirm the ranking suggested by the baseline phase. XGBoost achieved the highest overall performance (macro F1 = 82.17%, accuracy = 83.67%), with Logistic Regression establishing the linear ceiling at macro F1 = 75.57% - a meaningful gap that confirms the value of the non-linear ensemble approach for this task. The Decision Tree lagged considerably (macro F1 = 62.45%), reflecting its tendency to overfit in high-dimensional settings.

DIABLO's performance is worth brief comment. Despite being specifically designed for multi-omics integration, it did not surpass Logistic Regression and fell clearly behind XGBoost. This is partly attributable to the relatively small sample size ($n = 244$), which limits the ability of latent-variable models to reliably estimate cross-block covariance structure. More importantly for our purposes, DIABLO's sparse latent components offer a different form of interpretability than per-feature importance scores, giving us a different and less direct way to interpret the results. XGBoost, by contrast, provides gain-based feature importances at the level of individual features, which aligns precisely with the analytical objective of this work.

3.3.5 Two-Level miRNA Importance Validation Hypothesis

Before presenting the model results, it is important to take a step back and reflect that our primary question is not merely *which features help the classifier*, but rather *which features are genuinely informative for subtype discrimination*. Answering this question robustly requires more than a single source of evidence.

To this end, we propose the following hypothesis:

> *A miRNA (or clinical feature) is considered a* ***robustly validated discriminative marker*** *for a given breast cancer subtype if and only if it is independently selected by two complementary feature identification methods operating under fundamentally different assumptions: (i) the XGBoost internal feature importance, which reflects discriminative contribution within a trained multivariate model, and (ii) the ANOVA F-test, which reflects marginal statistical significance under the assumption of between-group mean differences.*

The rationale for this two-level approach stems directly from the complementary nature of the two methods:

- **ANOVA F-test (filter method)**: a model-agnostic statistical test that evaluates whether the means of a given feature differ significantly across subtype groups. A high F-score indicates that a feature is statistically distinguishable at the population level. Crucially, ANOVA operates on each feature in isolation, ignoring feature interactions. Features selected by ANOVA can therefore be interpreted as carrying *individually significant statistical signal* for subtype discrimination.

- **XGBoost gain importance (embedded method)**: the total gain attributed to a feature across all splits in the ensemble. Features with high gain are those that, in a multivariate discriminative context, contribute most to reducing classification error. This measure captures feature utility in the presence of all other features, including interactions and redundancies.

A feature appearing in *both* ranked lists is therefore supported by two independent confirmation levels: it is *statistically significant in isolation* and *discriminatively relevant in a multivariate learned model*. This double confirmation forms a particularly powerful filter for biomarker discovery: features verified by both methods are not only statistically significant in isolation but also deliver real, independent discrimination when evaluated inside the learning algorithm. Such convergence across distinct methodologies offers compelling support for prioritising these features as leading biomarker candidates. As shown in Section 3.3.6 and highlighted in Figure 3.9, doubly-validated features are consistently associated with stronger classification performance and robustness across subtypes, providing persuasive evidence that goes beyond what any single method could deliver alone, giving us confidence in the features identified by our approach and consider them strong candidates for further empirical validation.

3.3.6 XGBoost's Model Results

The optimised XGBoost model was evaluated under a binary One-vs-Rest (OvR) formulation: for each breast cancer subtype, a separate binary classifier was trained to distinguish patients of that subtype from all others. This One-vs-Rest strategy was adopted to enable per-subtype feature extraction (see Section 3.3.7) and to produce interpretable per-subtype performance metrics - after all, this models main purpose is to classify if a patient has a specific subtype or not.

Both feature configurations were evaluated: **miRNA only** (445 features) and **miRNA + clinical** (445 miRNA + 17 clinical features = 462 features).

3.3.6.1 Basal-Like

The Basal-like subtype yielded the highest classification performance across both configurations, with both miRNA-only and miRNA+clinical models achieving precision, recall, and F1-score of 1.00 on the test set. This result reflects the **strong and distinct miRNA expression signature** characteristic of Basal-like tumours, which are triple-negative, highly proliferative, and genomically distinct from the other subtypes.

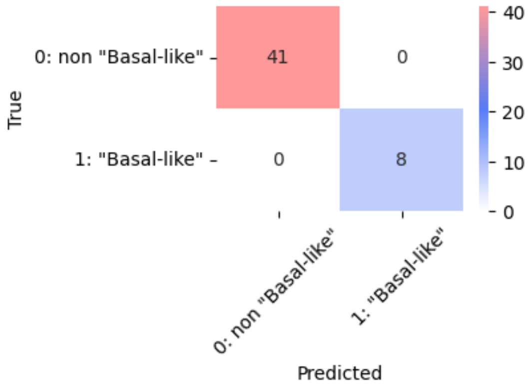
Table 3.9: Classification Performance for Basal-like Subtype

(a) Homogeneous Data

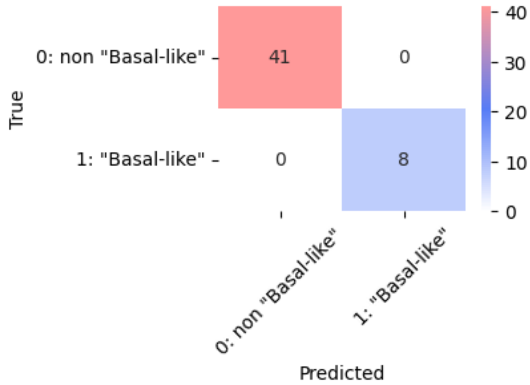
	prec.	rec.	f1	supp.
Not Basal	1.00	1.00	1.00	41
Basal	1.00	1.00	1.00	8
accuracy			1.00	49

(b) Heterogeneous Data

	prec.	rec.	f1	supp.
Not Basal	1.00	1.00	1.00	41
Basal	1.00	1.00	1.00	8
accuracy			1.00	49



(a)



(b)

Figure 3.4: Comparison of Classification Performance for Basal-like Breast Cancer Subtype classification where (a) has only miRNA data and (b) has both miRNA and clinical data.

3.3.6.2 HER2-Enriched

HER2-enriched classification proved to be more challenging. The miRNA-only model achieved an F1-score of 0.50 for the HER2-enriched class (precision 1.00, recall 0.33), indicating that while the model is highly precise - it does not produce false positives - it misses a substantial fraction of true HER2 cases (low recall). The macro-averaged F1-score across both classes was 0.73 (miRNA only) and remained similar with clinical data.

This is a known difficulty in the literature: HER2-enriched tumours exhibit considerable transcriptomic overlap with Luminal B, and their molecular distinctions are more reliably captured through protein expression or copy-number amplification of the *ERBB2* locus rather than through miRNA profiles alone. The addition of clinical features - particularly `HER2_STATUS` - provides meaningful supplementary signal for this subtype, as confirmed in the ANOVA analysis (Section 3.3.8).

Table 3.10: Classification Performance for HER2-enriched Subtype

(a) Homogeneous Data

	prec.	rec.	f1	supp.
Not HER2	0.91	1.00	0.96	43
HER2	1.00	0.33	0.50	6
accuracy			0.92	49

(b) Heterogeneous Data

	prec.	rec.	f1	supp.
Not HER2	0.91	1.00	0.96	43
HER2	1.00	0.33	0.50	6
accuracy			0.92	49

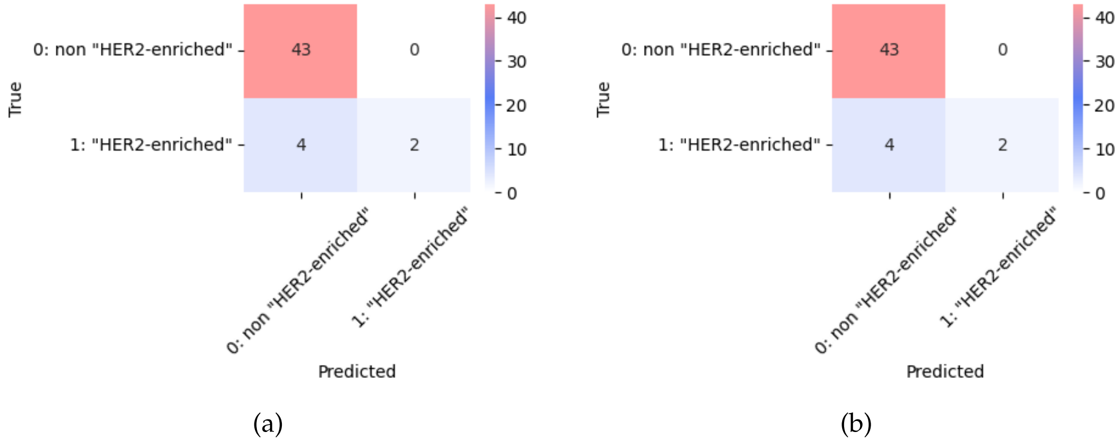


Figure 3.5: Comparison of Classification Performance for Her2-enriched Breast Cancer Subtype classification where (a) has only miRNA data and (b) has both miRNA and clinical data.

3.3.6.3 Luminal A

Luminal A classification yielded a weighted-average F1-score of 0.84 (miRNA only) and 0.82 (miRNA + clinical), with per-class F1-scores of 0.81 and 0.80, respectively. The slight decrease when adding clinical data warrants attention: it may reflect that clinical features introduce noise or redundancy for a subtype whose miRNA signature is already relatively distinct.

Table 3.11: Classification Performance for Luminal A Subtype

(a) Homogeneous Data

	prec.	rec.	f1	supp.
Not LumA	0.86	0.86	0.86	28
Luminal A	0.81	0.81	0.81	21
accuracy			0.84	49

(b) Heterogeneous Data

	prec.	rec.	f1	supp.
Not LumA	0.88	0.79	0.83	28
Luminal A	0.75	0.86	0.80	21
accuracy			0.82	49

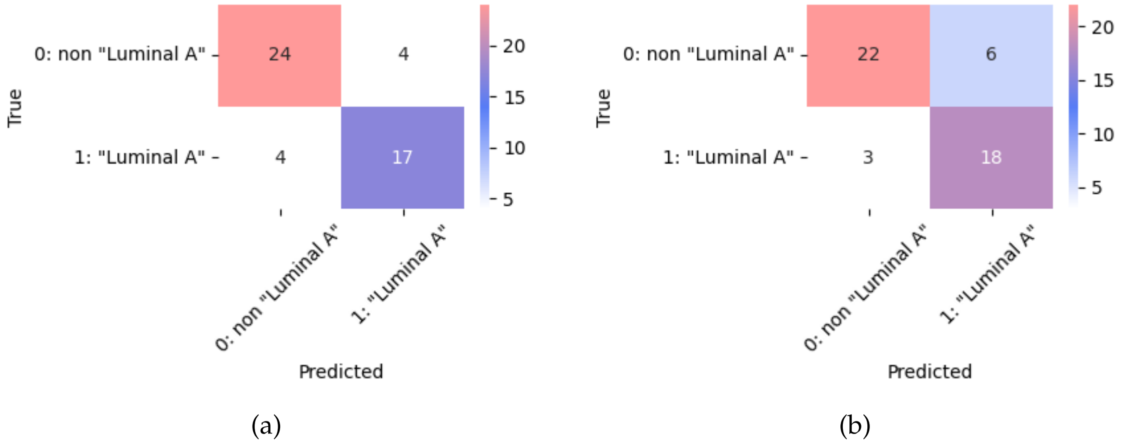


Figure 3.6: Comparison of Classification Performance for Luminal A Breast Cancer Subtype classification where (a) has only miRNA data and (b) has both miRNA and clinical data.

3.3.6.4 Luminal B

Luminal B was the most difficult subtype to classify, with an F1-score of 0.50 for the Luminal B class under the miRNA-only configuration (recall 0.36). With clinical data, the macro-averaged F1-score improved from 0.69 to 0.77, a notable gain suggesting that clinical covariates (particularly `ER_STATUS`) carry discriminative information not present in miRNA expression alone. This aligns with the known molecular heterogeneity of Luminal B, which overlaps substantially with Luminal A and lacks a clean miRNA-only discriminative boundary.

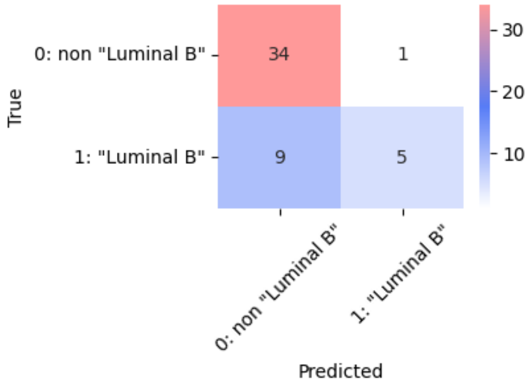
Table 3.12: Classification Performance for Luminal B Subtype

(a) Homogeneous Data

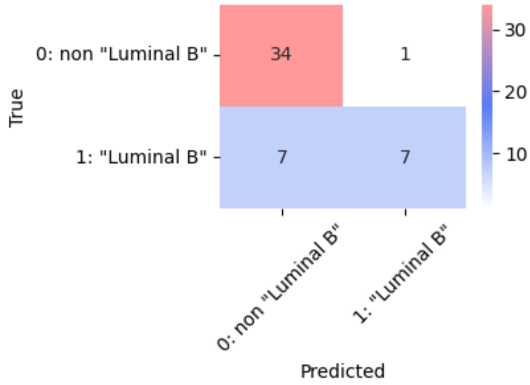
	prec.	rec.	f1	supp.
Not LumB	0.79	0.97	0.87	35
Luminal B	0.83	0.36	0.50	14
Accuracy			0.80	49

(b) Heterogeneous Data

	prec.	rec.	f1	supp.
Not LumB	0.83	0.97	0.89	35
Luminal B	0.88	0.50	0.64	14
Accuracy			0.84	49



(a)



(b)

Figure 3.7: Comparison of Classification Performance for Luminal B Breast Cancer Subtype classification where (a) has only miRNA data and (b) has both miRNA and clinical data.

3.3.7 Feature Extraction via XGBoost Booster

Having validated the classification performance of the optimised model, attention turns to the primary scientific objective: identifying the miRNAs most relevant to each subtype. To this end, the XGBoost internal booster was used to compute per-subtype **feature importances based on gain** - the average improvement in the loss function attributable to a feature across all splits in which it is used. This metric was chosen for evaluation because it allows us to understand the relative importance of each feature: a feature with substantially higher gain than its neighbours can be interpreted as carrying a disproportionate share of the discriminative signal for that subtype, whereas a more gradual decay in gain values suggests that the classification relies on a broader, more distributed set of features.

The binary OvR formulation is central here: by training a separate binary classifier for each subtype, the feature importances extracted are specific to the problem of distinguishing *that subtype from all others*, rather than reflecting an aggregate multiclass importance. This provides subtype-specific miRNA rankings.

Formally, for a given subtype $s \in \{\text{Basal-like}, \text{HER2-enriched}, \text{Luminal A}, \text{Luminal B}\}$, the binary label is defined as:

$$y_i^{(s)} = \begin{cases} 1 & \text{if patient } i \text{ belongs to subtype } s \\ 0 & \text{otherwise} \end{cases} \tag{3.2}$$

For each subtype and each feature configuration, the gain-based importance score was computed for every feature across all trees in the ensemble, and the top 20 features by cumulative gain were extracted and ranked.

Gain measures the total reduction in the model's objective function (loss) contributed by a feature across all trees. Unlike other metrics, it focuses on the *quality* of splits rather than their frequency:

Gain: The accumulated improvement in accuracy or error reduction provided by a feature.

Weight: The raw count of how many times a feature is used as a split.

Cover: The average number of samples or observations affected by a feature's splits.

While *weight* tells us how often a feature is used, **gain** tells us how much that feature actually matters to the final prediction - and this is exactly why we used the **gain** for our metric of feature importance selection.

Table 3.13 presents the top 20 miRNA features ranked by gain for each breast cancer subtype, under both the miRNA-only and miRNA + clinical configurations. For each subtype, the table reflects the features that the XGBoost model identified as most informative when learning to distinguish that subtype from all others, that is, the variables whose inclusion in the decision trees led to the greatest cumulative reduction in classification error.Reading across subtypes, the table thus provides a direct window into the molecular signatures that the model has learned to associate with each class.

Several patterns are noteworthy:

- **Basal-like**: MIR-577 dominates with ∼59% of total gain under miRNA-only and ∼57% with clinical data, suggesting it is a highly specific, singular discriminative signal. MIR-934 and MIR-135B are the next strongest contributors.
- **HER2-enriched**: The importance profile is more distributed (top feature MIR-30/30A accounts for only ∼15% of gain), consistent with the lower overall classification performance. With clinical data, `HER2_STATUS` enters the top 10, confirming the complementary role of the clinical receptor status.
- **Luminal A**: Gain is distributed across many features (top feature contributes only ∼5%), suggesting the absence of a dominant singular marker and a more diffuse miRNA signature for this subtype.
- **Luminal B**: Similarly distributed gain profile, with the top 20 features accounting for only ∼50% of total gain – the lowest concentration ratio across all subtypes, reflecting the heterogeneous nature of Luminal B.

Table 3.13: Top 20 XGBoost Feature Importance: miRNA Only vs. miRNA + Clinical Integration (% Gain)

Rank	Basal-like		HER2-enriched	
	miRNA Only	*miRNA + Clinical*	*miRNA Only*	*miRNA + Clinical*
1	MIR-577/577 (59.2%)	MIR-577/577 (56.9%)	MIR-30/30A (15.2%)	MIR-30/30A (10.7%)
2	MIR-135B/135B* (9.5%)	MIR-135B/135B* (10.5%)	MIR-34C/5P (7.5%)	MIR-3150B/3P (9.8%)
3	MIR-934/934 (8.5%)	MIR-934/934 (7.7%)	MIR-501/3P (6.9%)	MIR-34C/5P (8.4%)
4	MIR-190B/190B (4.3%)	**METHYLATION_CL.** (4.1%)	MIR-769/5P (6.2%)	MIR-769/5P (6.5%)
5	MIR-455/5P (2.8%)	MIR-190B/190B (3.8%)	MIR-192/192* (6.1%)	MIR-192/192* (5.4%)
6	MIR-584/584 (2.0%)	MIR-455/5P (3.4%)	MIR-3150B/3P (4.8%)	MIR-501/3P (4.7%)
7	MIR-505/505 (1.9%)	MIR-505/505 (1.9%)	MIR-187/187 (3.7%)	MIR-187/187 (4.3%)
8	MIR-19B-1/19B (1.9%)	MIR-2115/2115* (1.8%)	MIR-30C-2/2* (3.5%)	MIR-210/210 (3.1%)
9	MIR-2115/2115* (1.6%)	**SIGCLUST_MRNA** (1.6%)	MIR-142/5P (2.8%)	MIR-142/3P (3.0%)
10	MIR-9-1/9* (1.3%)	**CN_CLUSTER** (1.0%)	MIR-375/375 (2.7%)	**HER2_STATUS** (2.9%)
11	MIR-29/29A (1.0%)	MIR-516-1/5P (0.8%)	MIR-210/210 (2.6%)	MIR-497/497* (2.7%)
12	MIR-1296/1296 (0.9%)	MIR-9-1/9* (0.7%)	MIR-142/3P (2.6%)	MIR-184/184 (2.6%)
13	MIR-194-2/194* (0.7%)	MIR-455/3P (0.7%)	MIR-30C/3P (2.6%)	MIR-193/5P (2.5%)
14	MIR-501/3P (0.7%)	MIR-519A-1/519A* (0.6%)	MIR-2115/2115* (2.4%)	MIR-30C/3P (2.1%)
15	MIR-455/3P (0.6%)	MIR-24-2/2* (0.6%)	MIR-323B/3P (2.0%)	MIR-511/3P (2.1%)
16	MIR-675/675* (0.5%)	MIR-660/660 (0.6%)	MIR-301/301A (1.9%)	MIR-577/577 (2.0%)
17	MIR-744/744* (0.5%)	MIR-30C/3P (0.6%)	MIR-135-1/135A (1.9%)	MIR-651/651 (2.0%)
18	MIR-9-1/9 (0.5%)	MIR-20A/20A* (0.6%)	MIR-134/134 (1.8%)	MIR-181D/181D (1.9%)
19	MIR-18A/18A* (0.5%)	MIR-589/5P (0.5%)	MIR-217/217 (1.7%)	MIR-134/134 (1.9%)
20	MIR-519A-1/519A* (0.4%)	MIR-18A/18A* (0.5%)	MIR-184/184 (1.7%)	MIR-2115/2115* (1.9%)
Rank	**Luminal A**		**Luminal B**	
	miRNA Only	*miRNA + Clinical*	*miRNA Only*	*miRNA + Clinical*
1	MIR-30/30A (5.1%)	MIR-30/30A (5.4%)	MIR-3677/3P (5.1%)	MIR-190B/190B (5.5%)
2	MIR-501/3P (4.8%)	MIR-19A/19A (5.0%)	MIR-190B/190B (4.9%)	MIR-3677/3P (5.0%)
3	MIR-19A/19A (4.8%)	MIR-501/3P (4.9%)	MIR-126/126 (4.1%)	MIR-3615/3615 (3.8%)
4	MIR-1301/1301 (4.6%)	MIR-28/3P (4.2%)	MIR-3615/3615 (3.9%)	MIR-206/206 (3.7%)
5	MIR-15B/15B* (4.0%)	MIR-1301/1301 (4.1%)	MIR-1251/1251 (3.2%)	MIR-25/25 (3.5%)
6	MIR-28/3P (4.0%)	MIR-125A/3P (4.0%)	MIR-19B-1/1* (3.1%)	**SIGCLUST_MRNA** (3.4%)
7	MIR-125A/3P (3.8%)	MIR-15B/15B* (3.6%)	MIR-99/99A (2.8%)	MIR-1251/1251 (3.1%)
8	MIR-326/326 (3.5%)	MIR-106B/106B* (3.2%)	MIR-206/206 (2.2%)	MIR-338/3P (2.8%)
9	MIR-105-1/105 (3.3%)	MIR-205/205* (3.1%)	MIR-101-1/101 (1.9%)	MIR-205/205* (2.4%)
10	MIR-664/664 (3.1%)	MIR-326/326 (2.5%)	MIR-30D/30D (1.9%)	MIR-126/126 (2.0%)
11	MIR-3200/3P (2.8%)	MIR-145/145 (2.4%)	MIR-4326/4326 (1.8%)	MIR-99/99A (1.9%)
12	MIR-205/205* (2.4%)	MIR-105-1/105 (2.4%)	MIR-25/25 (1.8%)	MIR-30D/30D (1.9%)
13	MIR-222/222 (1.9%)	MIR-451/451 (2.0%)	LET-7E/7E* (1.8%)	MIR-944/944 (1.7%)
14	MIR-451/451 (1.8%)	MIR-31/31* (1.9%)	MIR-24-1/24 (1.8%)	MIR-454/454 (1.7%)
15	MIR-106B/106B* (1.6%)	**CN_CLUSTER** (1.9%)	MIR-15A/15A* (1.7%)	MIR-24-1/24 (1.6%)
16	MIR-3614/5P (1.6%)	MIR-642/642A (1.9%)	MIR-20B/20B* (1.6%)	MIR-15A/15A* (1.5%)
17	MIR-101-1/101 (1.5%)	MIR-3200/3P (1.8%)	MIR-15B/15B (1.6%)	MIR-20B/20B* (1.5%)
18	MIR-32/32* (1.5%)	MIR-664/664 (1.7%)	MIR-874/874 (1.6%)	MIR-874/874 (1.5%)
19	MIR-200A/200A (1.4%)	MIR-345/345 (1.4%)	MIR-484/484 (1.5%)	MIR-582/3P (1.5%)
20	MIR-425/425* (1.4%)	MIR-190B/190B (1.3%)	MIR-205/205* (1.5%)	MIR-4326/4326 (1.5%)

3.3.8 Statistical Corroboration via ANOVA F-Test

The feature importances derived from XGBoost in the previous section are, by their nature, *model-dependent*: they reflect the contribution of each feature within a specific learned ensemble, under a specific training regime. The two concepts (discriminative utility and statistical significance) are related but not equivalent, and conflating them would weaken any biological conclusions drawn from the analysis.

This is the fundamental motivation for introducing the ANOVA F-test at this stage. Unlike XGBoost, ANOVA is entirely model-agnostic: it evaluates each feature independently of any classifier, asking a direct statistical question: **does the mean expression of this miRNA differ significantly between patients of a given subtype and all others?** A feature that ranks highly under ANOVA is not "important to a model"; it is **statistically distinguishable at the population level**, under the well-defined assumptions of analysis of variance. This distinction is what gives the ANOVA results their evidential value here: they provide a source of confirmation that operates through an entirely different lens.

When a miRNA is selected as important by XGBoost *and* identified as statistically significant by ANOVA, the two confirmations are independent. The model did not influence the F-scores, and the F-scores did not influence the model. It is precisely this independence that makes the intersection meaningful: a feature that survives both filters is not an artefact of one method's assumptions or luck, but a signal robust enough to emerge from two fundamentally different analytical paradigms.

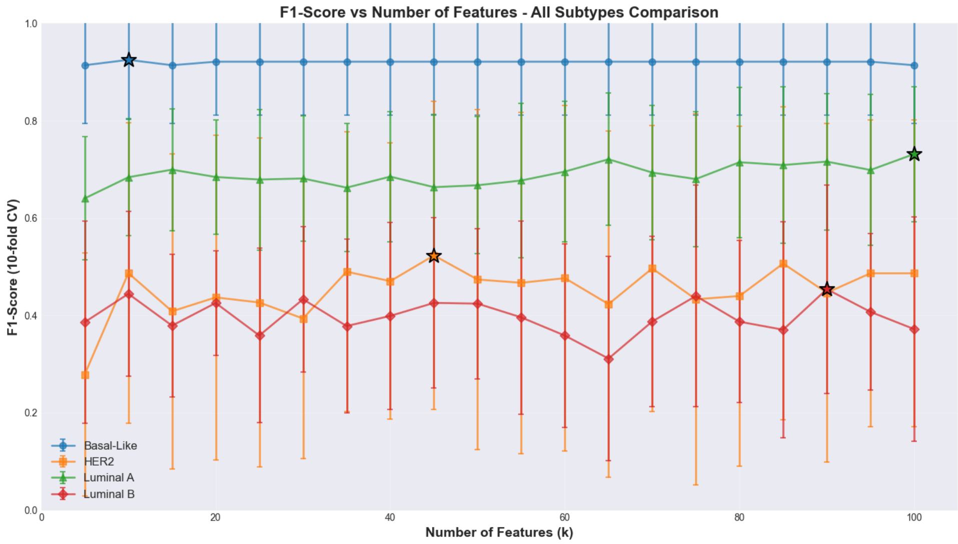
Given this, for each subtype $s \in \{\text{Basal-like, HER2-enriched, Luminal A, Luminal B}\}$, the analysis was framed as a binary OvR problem, consistent with the XGBoost experiments. The F-statistic for each feature j is defined as the ratio of between-group to within-group variance:

$$F_j = \frac{n_1\,(\bar{x}_{j,1} - \bar{x}_j)^2 \;+\; n_0\,(\bar{x}_{j,0} - \bar{x}_j)^2}{\displaystyle\sum_{k\in\{0,1\}} \sum_{i:\,y_i^{(s)}=k} (x_{ij} - \bar{x}_{j,k})^2 \,/\, (n-2)} \tag{3.3}$$

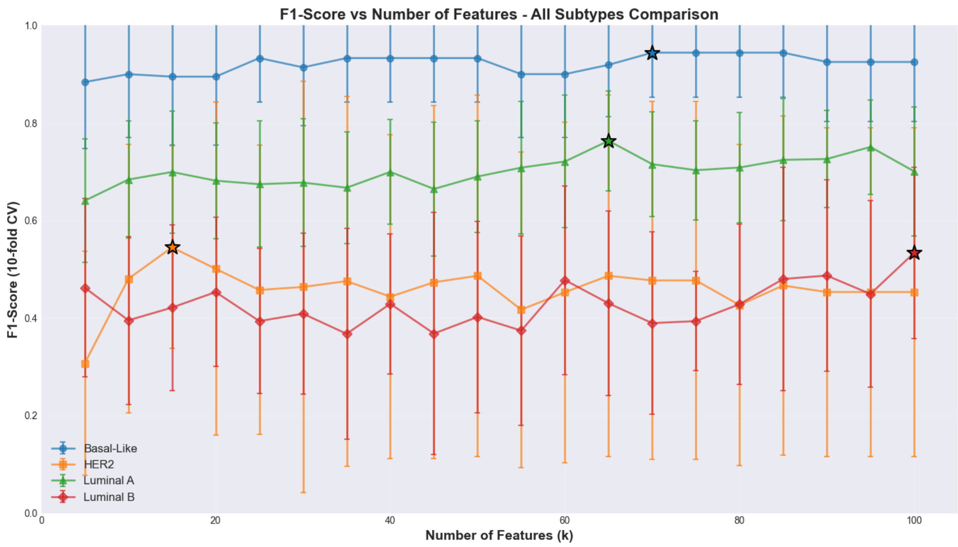
where n_1 and n_0 are the sizes of the positive and negative groups, $\bar{x}_{j,k}$ is the mean expression of feature j in group k, and $\bar{x}_j$ is the overall mean. A large F_j indicates that the expression of feature j varies more *between* the two groups than it does *within* them - the defining criterion of a statistically discriminative marker.

All features were standardised via z-score normalisation prior to computing F-scores, ensuring that differences in expression scale across miRNAs do not artificially inflate or deflate individual F-statistics. Feature selection was embedded in a cross-validation pipeline with a downstream XGBoost classifier, ensuring that feature selection was performed exclusively on training data at each fold, with no information leakage from the test set. The optimal number of features k was determined per subtype as the value maximising the cross-validated macro F1-score. Additionally, features were added incrementally in decreasing order of F-score, and the resulting classification performance trajectory was recorded: revealing both *how many* features are needed and *which individual features* contribute most to the discriminative signal.

The results reveal a striking gradient across subtypes **that directly mirrors the XGBoost findings**. For Basal-like, classification performance saturates rapidly at around $k \approx 10$ (F1 ≈ 0.92), with MIR-934, MIR-577, and MIR-135B alone accounting for the bulk



(a) miRNA only



(b) miRNA + clinical

Figure 3.8: F1-score (10-fold CV) as a function of the number of features k for each breast cancer subtype, under two feature configurations. Stars indicate the optimal k selected for each subtype.

of the gain, their F-scores of 386, 358, and 292 respectively are an order of magnitude above those of any other subtype's top features, reflecting an exceptionally strong and concentrated statistical signal. For HER2-enriched, by contrast, performance continues to improve across the full tested range of $k \leq 50$, and the top F-scores barely exceed 50 - confirming, through a completely independent lens, what XGBoost already suggested: this subtype lacks a small dominant set of miRNA markers. Luminal A reaches its performance ceiling around $k = 20$ (F1 ≈ 0.73), while Luminal B exhibits the weakest and most volatile trajectory of all, with top F-scores around 34 - further evidence that its molecular boundary is diffuse and that no individual miRNA cleanly separates it from adjacent subtypes.

The consistency of this picture across two independent methods is not incidental. It is the convergence this work relies upon to move from model-derived feature rankings to biologically grounded candidate miRNA panels, as formalised in the following section.

With both the XGBoost gain importances and the ANOVA F-scores computed independently, we now bring them together to identify the **doubly-validated miRNA panels** for each subtype, as formalised in Section 3.3.5.

Figure 3.9 presents, for each subtype and each feature configuration, the features that appear in both the ANOVA top-ranked list (at the optimal k) and the XGBoost top-20 gain ranking.

3.3.9 Summary of Doubly-Validated Features

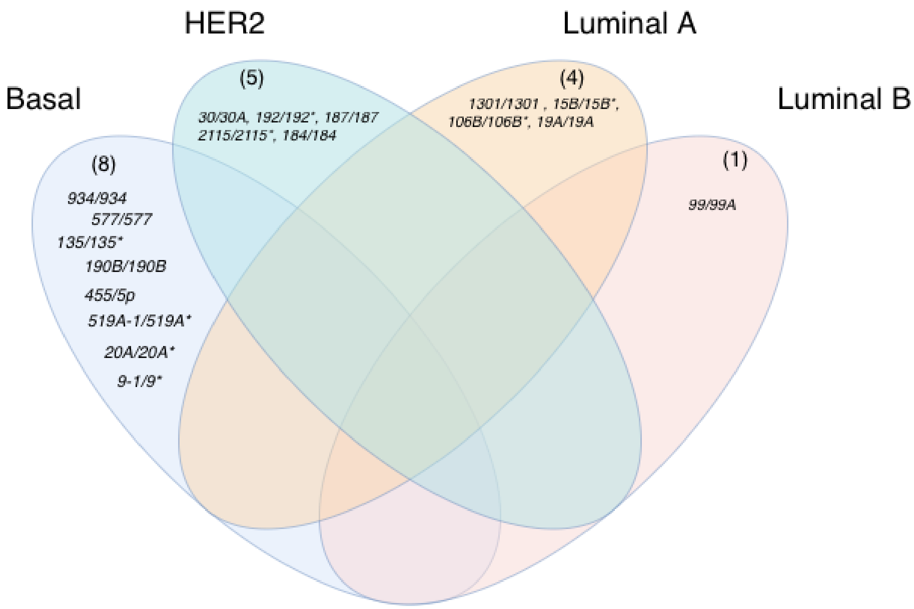
The number of doubly-validated features per subtype and condition is summarised below:

Subtype	**miRNA only**	**miRNA + clinical**
Basal-like	8	8
HER2-enriched	5	7
Luminal A	4	4
Luminal B	1	4

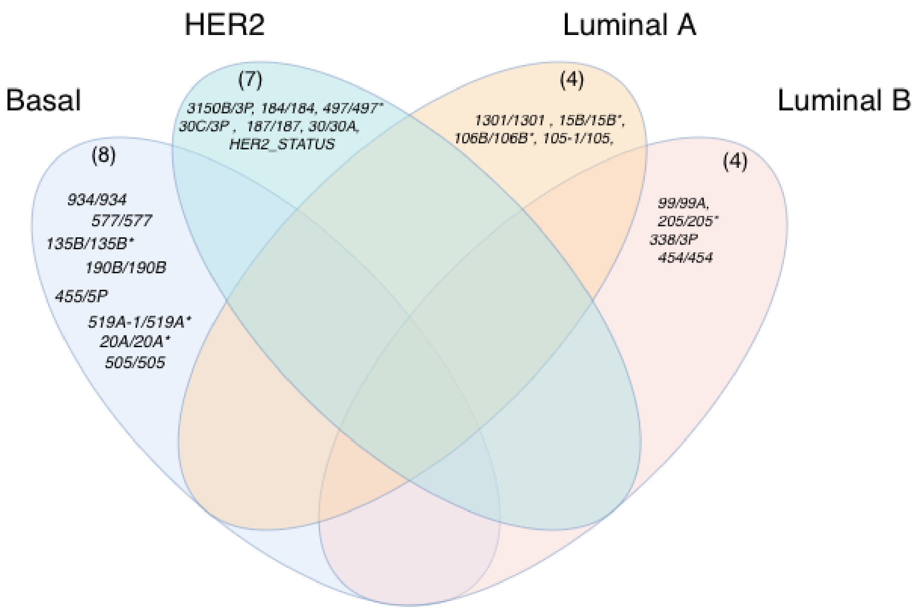
The consistent increase in validated features when clinical data is added - especially for HER2-enriched (+2) and Luminal B (+3) - is itself a finding: it quantifies the contribution of clinical covariates not just to classification accuracy, but to the identifiability of subtype-specific molecular markers.

Basal-Like The three features **MIR-577, MIR-934, and MIR-135B** are doubly validated with the highest confidence: they rank in the top 3 in both ANOVA (F-scores of 358, 386, and 292 respectively) and XGBoost gain. MIR-577's dominance in XGBoost ($\sim 59\%$ gain) paired with its high ANOVA F-score positions it as the **strongest candidate discriminative marker for Basal-like**. MIR-190B, MIR-455-5p, and MIR-505 complete the core doubly-validated panel. With clinical data, MIR-519A and MIR-20A enter the intersection.

HER2-Enriched Without clinical data, only 5 features are doubly validated, reflecting the difficulty of this subtype. With clinical data, **HER2_STATUS** tops the ANOVA ranking (F-score 76.90) and also appears in the XGBoost top 10, making it the most strongly



(a) miRNA only



(b) miRNA and Clinical data

Note: The prefix "MIR-" is omitted for brevity and to improve readability.

Figure 3.9: Feature intersections showing doubly-validated markers for all subtypes under both miRNA-only and miRNA+clinical conditions.

validated feature for this subtype - and the only clinical feature to achieve double validation. This finding strongly supports the clinical intuition that HER2 receptor status is the defining feature of this subtype. Among miRNAs, **MIR-30/30A and MIR-3150B-3p** are the most robustly validated, appearing in both lists under both conditions.

Luminal A The four doubly-validated features without clinical data - **MIR-1301, MIR-15B, MIR-19A, and MIR-106B** - form a compact panel with moderate but consistent signal. A notable discrepancy is observed: **MIR-942** achieves the highest ANOVA F-score for this subtype (102.90) and generates the largest single-feature F1 improvement (+0.56), yet does not appear in the XGBoost top 20. This may indicate that MIR-942's discriminative power is primarily marginal - it differs strongly in mean expression between Luminal A and other subtypes - but in the presence of other features within the model, its independent contribution is reduced (possibly due to correlation with other included features). This discrepancy is itself an informative finding: it illustrates the complementarity of the two methods and cautions against relying exclusively on either one.

Luminal B Luminal B presents the weakest intersection, with only 1 doubly-validated feature (**MIR-99/99A**) under the miRNA-only condition. The overall low ANOVA F-scores (top feature F-score ≈ 34, compared to 386 for Basal-like) indicate that no single miRNA differs dramatically in mean expression between Luminal B and other subtypes. With clinical data, 4 features are validated, including **MIR-99/99A** as the most consistently appearing feature across both methods and both conditions - making it the strongest candidate marker for Luminal B.

3.3.10 Limitations and Caveats

Several limitations of this analysis deserve explicit acknowledgement:

1. The dataset comprises only 244 samples, which limits statistical power - particularly for minority subtypes (HER2-enriched: $n = 29$). Validation on a larger, independent cohort is necessary before the identified panels can be considered clinically actionable.

2. The two-level validation framework is confirmatory, not exploratory: it increases confidence in features already appearing in both ranked lists, but cannot rule out relevant features that are highly correlated with, and therefore displaced by, other selected features (redundancy effects, especially in XGBoost gain).

3. ANOVA's assumptions (normality, homoscedasticity) may not hold strictly for all miRNA features; however, the large sample sizes within each group provide some robustness via the central limit theorem.

Despite these limitations, the **convergence of two fundamentally different methods** onto a consistent set of features, under both data configurations and across a subtype as genomically distinct as Basal-like, provides a degree of confidence in the identified panels that neither method alone could establish.

3.4 Group-Based and Unsupervised Subtyping

The previous section addressed the problem of breast cancer subtype classification through a supervised learning lens: labelled samples were used to train models that learn an explicit decision boundary between the four PAM50 subtypes. While this is the conventional formulation for clinical classification tasks, it rests on a fundamental assumption that the labels are reliable, complete, and the only relevant structure in the data.

This section sets those labels aside, thus transforming the core question we ask: **does the miRNA expression data itself, without any guidance from clinical annotations, organise into groups that reflect the known biological subtypes?** If such structure exists, it suggests that the subtype-discriminating signal is intrinsic to the transcriptomic landscape and not merely an artefact of how patients were clinically classified. Moreover, for any cluster that aligns strongly with a known subtype, we can identify which miRNAs drive that alignment and complement the analysis of the supervised models.

3.4.1 Why This Approach Is Non-Conventional

Applying unsupervised learning to a problem that has well-defined class labels is, by most standards, a non-conventional choice. Standard practice in breast cancer subtype research, or in other cancer research areas, would typically involve using the labels (in this case the PAM50 gene signature) to train a supervised classifier and then treat subtype assignment as a supervised classification problem leveraging the ground truth (similiar to [8]).

The unconventional nature of this approach is deliberate. Rather than asking **"can we predict the subtype of a patient?"**, we ask **"does the data naturally group itself in a way that corresponds to known subtypes?"**. These are fundamentally different questions. The first is a performance question; the second is a structural one. A high-performing supervised classifier may exploit subtle decision boundaries that have no clear biological interpretation. A cluster that achieves high purity with respect to a known subtype, by contrast, asserts that the subtype occupies a geometrically coherent and isolated region of the miRNA expression space, making it a stronger biological claim.

Furthermore, this framing makes the analysis robust to label noise: the clustering algorithm never sees the labels, so any correspondence between clusters and subtypes that emerges is entirely data-driven.

3.4.2 Implementation

3.4.2.1 Dataset and Experimental Conditions

Experiments were conducted on the small cohort ($n = 244$ samples). Two experimental conditions were evaluated in parallel:

- **miRNA only:** 442 miRNA expression features, after preprocessing.
- **miRNA + clinical:** 462 features, combining miRNA expression with clinical covariates.

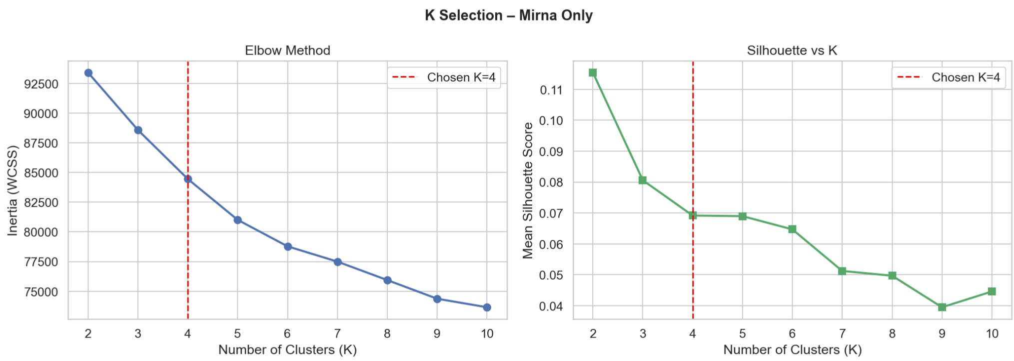
Running both conditions through the same pipeline serves a concrete purpose: it allows us to assess whether miRNA expression alone carries sufficient structure to recover biologically meaningful clusters, and whether clinical data strengthens or dilutes that signal. All conclusions are drawn in parallel for both conditions so that differences can be attributed to the feature set rather than to any other methodological choice.

While K-Means minimises Euclidean distances in feature space, making it sensitive to feature scale. A feature with values in the range $[0, 1000]$ will dominate distance calculations over a feature in $[0, 1]$, regardless of their relative biological importance. Accordingly, all features were standardised using *StandardScaler*, transforming each feature to zero mean and unit variance prior to clustering.

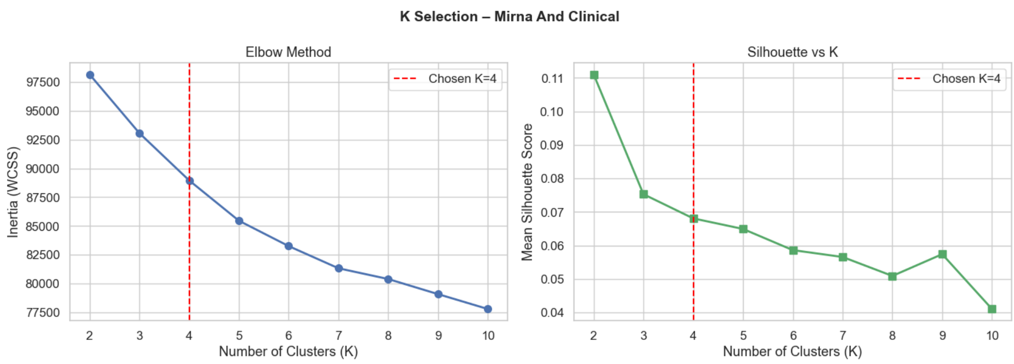
No correlation-based redundancy removal was applied. Although correlated features may inflate the weight of certain directions in Euclidean space, removing them based on pairwise correlation would risk discarding biologically relevant miRNAs which, in our case, we cannot afford. This would mean that a miRNA identified as a discriminative feature in the supervised analysis could be silently removed as a correlated redundant, breaking the cross-comparison between the two approaches. Retaining the full feature space ensures that the post-hoc feature importance analysis (Section 3.4.3.3) operates on the same miRNA vocabulary as the supervised section, enabling a direct comparison of which miRNAs are flagged by each paradigm. The presence of correlated features is acknowledged as a methodological constraint.

3.4.2.2 Number of clusters

The number of clusters K was set to 4, corresponding to the four known PAM50 breast cancer subtypes (Basal-Like, HER2-Enriched, Luminal A, and Luminal B): since the ultimate goal is to assess whether the unsupervised structure aligns with known biology, fixing K to match the number of subtypes is the only choice that makes the external evaluation metrics meaningful. Figure 3.10 shows the diagnostic curves for both conditions. In both cases, the elbow method produces a gradual, continuously declining inertia curve with no discernible inflection point at any K. The silhouette score peaks at $K = 2$ (≈ 0.115 for miRNA-only; ≈ 0.111 for miRNA + clinical) and decreases monotonically thereafter, reaching ≈ 0.069 at $K = 4$ in both conditions. The two curves are strikingly similar, indicating that neither feature set produces a geometry that the data-driven criteria can resolve into cleanly separated clusters. Consequently, neither curve provides unambiguous support for $K = 4$, reinforcing that the choice must be grounded in domain knowledge.



(a) miRNA-only condition.



(b) miRNA and clinical data condition.

Figure 3.10: Elbow method (inertia vs. K, left) and mean silhouette score vs. K (right) for both experimental conditions. The red dashed line marks the chosen value $K = 4$. Both conditions produce curves of the same shape, with no clear elbow and a silhouette peak at $K = 2$.

This choice is justified on two grounds. First, purity evaluated against the four PAM50 classes is only meaningful when the number of clusters matches the number of classes: setting $K = 3$ or $K = 5$ would inevitably create clusters that either merge or split subtypes, making any purity comparison with the supervised results ambiguous. Second, one of the central findings of the supervised section was that HER2-enriched and Luminal B are the hardest subtypes to separate. Setting $K = 4$ preserves that boundary, allowing us to directly test whether K-Means, operating without labels, can recover it from expression data alone.

3.4.2.3 Model Fitting

K-Means was fitted using the K-Means++ initialisation strategy, which selects initial centroids with probability proportional to their squared distance from already-chosen centroids. Compared to purely random initialisation, this approach reduces the risk of converging to poor local minima and generally accelerates convergence, making it particularly advantageous in high-dimensional spaces where the inertia landscape contains numerous local optima.

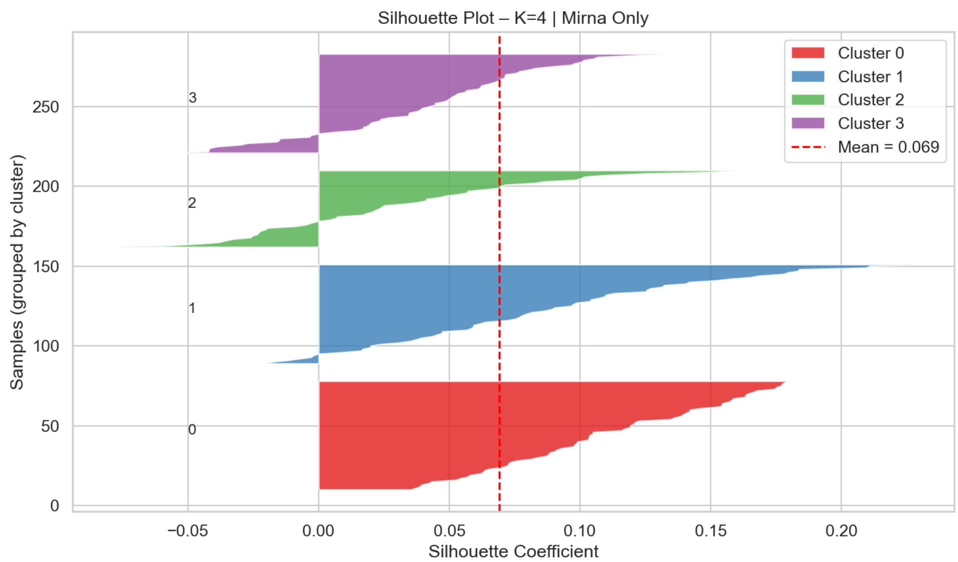
To further mitigate sensitivity to initialisation, 10 independent random restarts were performed, each starting from a different K-Means++ initialised set of centroids. The solution yielding the lowest inertia across all restarts was retained as the final model. Convergence within each restart was assessed by monitoring centroid displacement between successive iterations, with a fixed tolerance threshold below which the algorithm was considered to have converged. A fixed random seed was set throughout to ensure full reproducibility of both the initialisation and the restart sequence.

Figure 3.11 shows the per-sample silhouette plots at $K = 4$ for both conditions. The mean silhouette score is low in both cases (0.069 for miRNA-only; 0.068 for miRNA + clinical), consistent with the absence of a clear elbow and confirming that the four clusters do not form geometrically compact, well-separated regions in the standardised feature space. This is not unexpected: K-Means assumes convex, isotropic clusters of similar size, an assumption that high-dimensional transcriptomic data rarely satisfies. The low silhouette is therefore a property of the data geometry, not necessarily evidence that no biological structure exists.

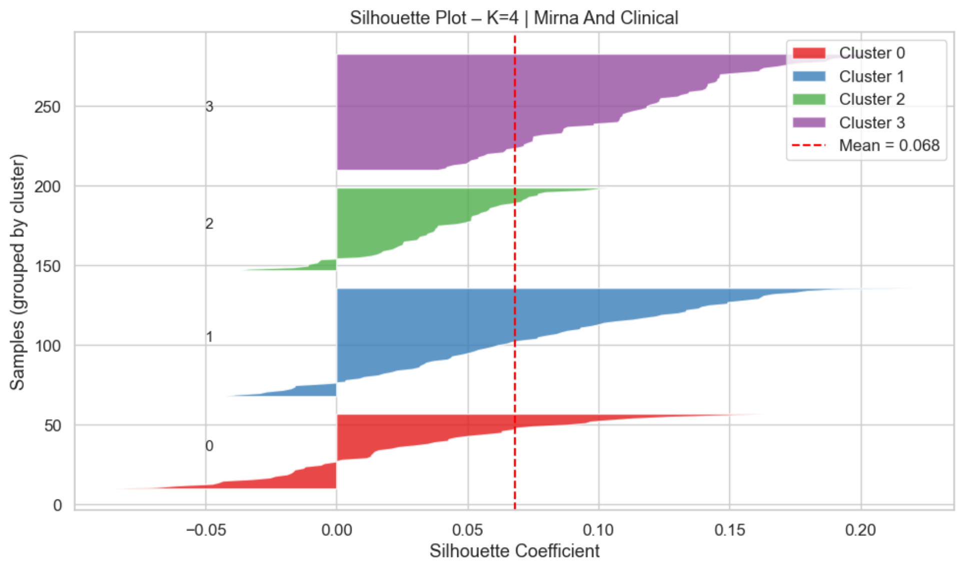
Despite this similarity in aggregate score, the two conditions differ in one important respect. In the miRNA-only setting, all four clusters produce predominantly positive silhouette coefficients with broadly uniform band widths, indicating that samples are consistently assigned to their nearest centroid. In the miRNA + clinical condition, Cluster 0 exhibits a substantial proportion of negative coefficients, suggesting that a non-trivial number of samples are geometrically closer to a neighbouring cluster than to their own. This asymmetry is an early indication that the addition of clinical features introduces heterogeneity that K-Means cannot resolve at $K = 4$, a hypothesis that will be confirmed by the external evaluation metrics in the following section.

3.4.3 Clustering Analysis and Evaluation

Having fitted the model and characterised the geometry of the resulting partition, we now turn to the central question of this section: does K-Means, operating entirely without label information, recover a partition that corresponds to the known PAM50 subtype structure? And does the addition of clinical covariates strengthen or weaken that correspondence? We address both questions at three levels of increasing granularity: aggregate evaluation metrics, per-cluster composition, and miRNA feature signatures. Both experimental



(a) miRNA-only condition.



(b) miRNA and clinical condition.

Figure 3.11: Per-sample silhouette plots for $K = 4$ under both experimental conditions. Each coloured band corresponds to one cluster. Bars to the right of zero indicate well-assigned samples; bars to the left indicate potential misassignments. The red dashed line marks the mean silhouette score (0.069 for miRNA-only and 0.068 for miRNA + clinical). Note Cluster 0 in the miRNA + clinical condition: it is the only cluster in either condition to exhibit a substantial fraction of negative coefficients.

conditions are analysed in parallel throughout, so that differences can be attributed unambiguously to the effect of the feature set.

3.4.3.1 Cluster Quality Metrics

The K-Means algorithm was evaluated under two experimental conditions - miRNA expression features alone (442 features) and miRNA expression combined with clinical covariates (462 features) - both fitted on the small cohort of $n = 244$ samples with $K = 4$ clusters, in correspondence with the four PAM50 breast cancer subtypes. Given this, Table 3.14 summarises the four evaluation metrics across both conditions.

Table 3.14: K-Means performance summary for $K = 4$ on the small cohort ($n = 244$).

Condition	Features	Silhouette	Purity	ARI	AMI
miRNA only	442	0.069	0.611	0.203	0.258
miRNA + clinical	462	0.068	0.598	0.192	0.222

The external metrics tell an informative story. An overall purity of 0.611 for the miRNA-only condition means that 61% of samples are assigned to a cluster where their true PAM50 subtype is the majority class. Under random assignment of 244 samples into 4 equal-sized clusters, the expected purity would be approximately 0.25; the observed value therefore represents a meaningful above-chance alignment between the discovered partition and the ground-truth labels. The ARI of 0.203 and AMI of 0.258 are similarly modest but non-trivial, both well above the zero baseline expected from a random partition.

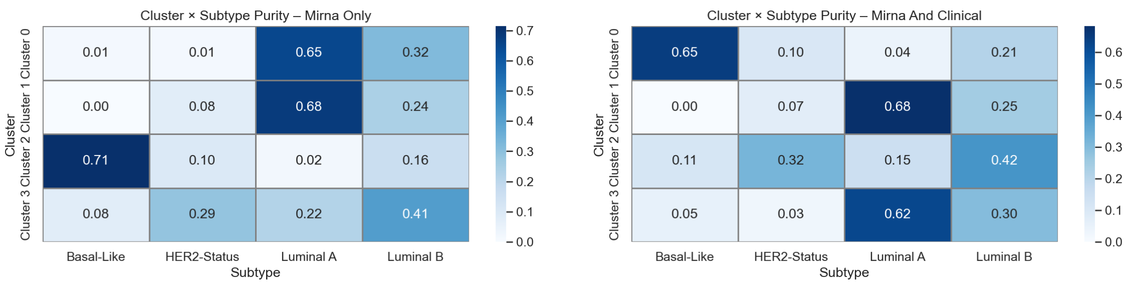
Across all four metrics, the miRNA-only condition consistently outperforms miRNA + clinical. The differences are small in absolute terms but systematic in direction, with the AMI showing the largest relative drop ($\approx 14\%$) when clinical features are added. This consistency points to a clear conclusion: the subtype-discriminating signal in this cohort resides primarily in the miRNA expression space, and appending clinical covariates introduces variation that is orthogonal to PAM50 subtype boundaries. Because K-Means treats all standardised features symmetrically, each clinical variable occupies one dimension with the same weight as any miRNA, effectively adding independent axes of variation that pull centroids away from the miRNA-driven structure and dilute the clustering signal.

3.4.3.2 Cluster Composition

While aggregate metrics provide a global picture, they do not reveal which subtypes are correctly isolated and which are conflated. Examining the internal composition of each cluster answers that question and allows us to identify which PAM50 subtypes produce a geometrically coherent signal in expression space and which are systematically confused with one another. An important methodological note before proceeding: K-Means cluster indices are arbitrary and not comparable across conditions by number alone; two clusters are considered to correspond to the same biological entity when they share the same dominant PAM50 subtype. Table 3.15 and Figure 3.12 present the full composition for both conditions, while Figures 3.13 and 3.14 show the subtype distribution within each cluster.

Table 3.15: Cluster composition for both experimental conditions ($K = 4$, $n = 244$). Each cell reports the raw count of samples of that subtype in that cluster. The dominant subtype per cluster is indicated in the rightmost column with its within-cluster purity in parentheses.

Condition	Cluster	Basal	HER2	Lum. A	Lum. B	Dominant (purity)
miRNA only	C0 ($n = 68$)	0	6	**45**	17	Lum. A (66.2%)
	C1 ($n = 47$)	**34**	5	1	7	Basal-Like (72.3%)
	C2 ($n = 66$)	3	0	**42**	21	Lum. A (63.6%)
	C3 ($n = 63$)	4	18	15	**26**	Lum. B (41.3%)
miRNA + clinical	C0 ($n = 62$)	0	4	**43**	15	Lum. A (69.4%)
	C1 ($n = 65$)	5	18	15	**27**	Lum. B (41.5%)
	C2 ($n = 67$)	2	2	**43**	20	Lum. A (64.2%)
	C3 ($n = 50$)	**34**	5	2	9	Basal-Like (68.0%)



(a) miRNA only

(b) miRNA + clinical

Figure 3.12: Cluster × Subtype purity heatmaps for both conditions. Each cell reports the proportion of samples of that subtype within the cluster. Note that cluster indices are not directly comparable across conditions; the same biological structure may appear under a different index in each panel.

At the level of dominant subtypes, the partition recovered by both conditions is strikingly similar: both produce one Basal-Like-dominant cluster, two Luminal A-dominant clusters, and one mixed Luminal B / HER2-Enriched cluster. This structural consistency is itself a meaningful result: it suggests that the four-cluster organisation of the data is driven primarily by the miRNA expression component, and that the addition of clinical features does not fundamentally alter the topology of the recovered partition. The differences between conditions are therefore quantitative rather than qualitative.

The Basal-Like cluster is the most biologically coherent in both conditions. In the miRNA-only condition, the Basal-Like cluster (C1) achieves a purity of 72.3%, with 34 out of 47 samples belonging to the Basal-Like subtype. Under the miRNA + clinical condition, the corresponding cluster (C3) recovers the same 34 Basal-Like samples but draws in 3 additional non-Basal samples, reducing purity slightly to 68.0% ($n = 50$). The high coherence of this cluster in both conditions is consistent with the clinical literature: Basal-Like (triple-negative) tumours are known to exhibit a markedly distinct molecular profile from hormone-receptor-positive subtypes [48], and this distinction is clearly captured by miRNA expression alone. The small purity drop when clinical features are added is a direct consequence of the metric dilution described above: a few non-Basal samples whose clinical profile places them geometrically close to the Basal centroid are drawn into the cluster.

Luminal A is fragmented into two clusters in both conditions. Both conditions produce two distinct Luminal A-dominant clusters rather than a single cohesive one. In the miRNA-only case these are C0 (66.2% purity, $n = 68$) and C2 (63.6% purity, $n = 66$); under miRNA + clinical they are C0 (69.4% purity, $n = 62$) and C2 (64.2% purity, $n = 67$). Both clusters in both conditions show a secondary Luminal B component and negligible Basal-Like or HER2-Enriched representation. Rather than being a failure of the clustering, this fragmentation is informative in itself: Luminal A tumours do not occupy a single compact region in miRNA expression space but form at least two separable subgroups with distinct centroid profiles. This is consistent with the supervised analysis, where Luminal A showed heterogeneous behaviour across feature subsets, and aligns with evidence in the literature of biological heterogeneity within the Luminal A class [48].

Luminal B and HER2-Enriched remain systematically mixed in both conditions. Both conditions produce one cluster in which Luminal B and HER2-Enriched samples co-dominate, with near-identical composition: 41.3% Luminal B and 28.6% HER2-Enriched in the miRNA-only condition; 41.5% Luminal B and 27.7% HER2-Enriched in the miRNA + clinical condition. This near-identical mixing across both conditions confirms that the difficulty of separating these two subtypes is an intrinsic property of the miRNA expression space, not an artefact of the feature set, directly echoing the confusion observed in the supervised analysis.

3.4.3.3 Feature Signatures

Knowing that each cluster aligns to a dominant PAM50 subtype is only part of the story. The next question is which miRNAs define each cluster's centroid: that is, which features deviate most strongly from the global dataset mean within each group. These are the miRNAs that the unsupervised algorithm implicitly used to separate the data, and identifying them serves two purposes. First, it provides a biologically interpretable signature for each cluster. Second, and more importantly in the context of this thesis, it enables a direct cross-comparison with the discriminative features identified in the supervised

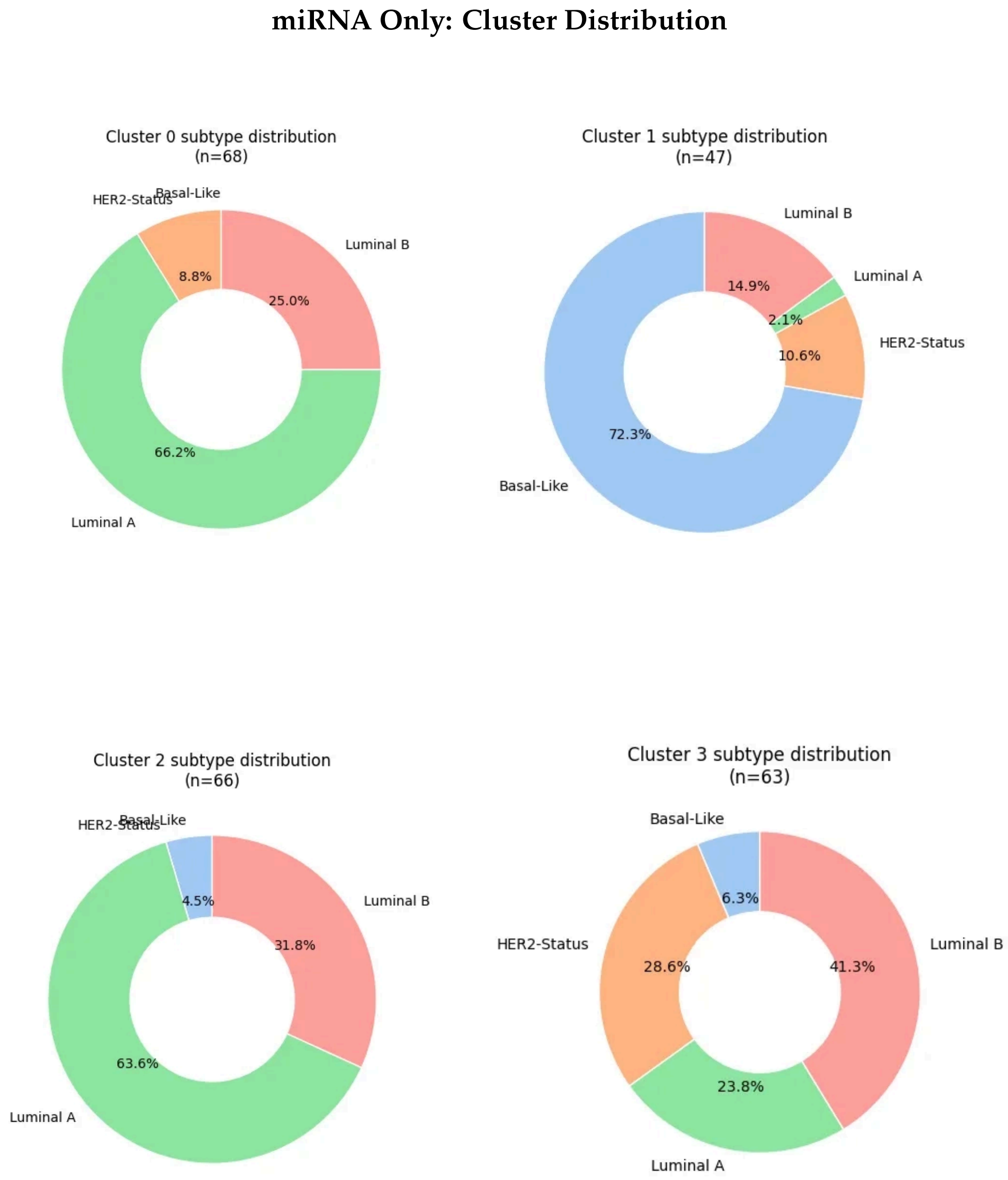


Figure 3.13: Breast Cancer Subtype Distribution: miRNA Only Dataset.

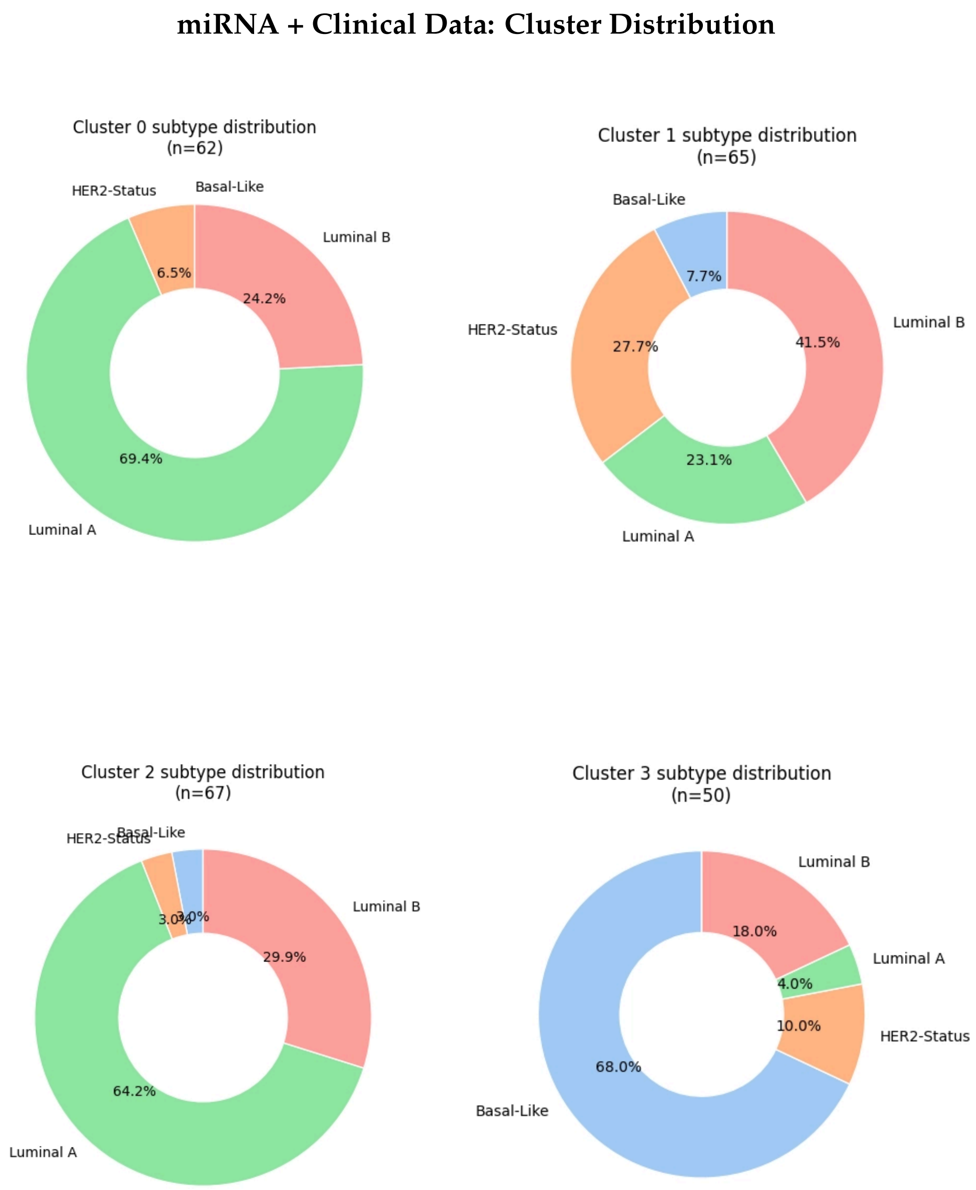


Figure 3.14: Breast Cancer Subtype Distribution: miRNA + Clinical Dataset.

section, allowing us to assess whether both paradigms converge on the same miRNAs or capture complementary aspects of subtype biology. Both conditions are examined here so that the stability of the identified signatures can be directly assessed.

Each miRNA is ranked by the absolute deviation of its cluster-level mean expression from the global cohort mean, expressed in standard deviation units (absolute centroid z-score). A miRNA that is consistently over or underexpressed in a given cluster relative to the full cohort receives a high absolute score, marking it as characteristic of that cluster. This is the canonical way to interpret K-Means centroids, since the centroid is simply the mean expression profile of the cluster [26].

The figures 3.15 and 3.16 show the top-20 feature rankings for all four clusters under both conditions (with miRNA-only and miRNA+clinical features), and table 3.16 summarises the top-5 features per biological cluster type.

Table 3.16: Top 5 miRNA features per biological cluster type by absolute centroid z-score, for both experimental conditions. Clusters are identified by dominant PAM50 subtype rather than by index, since K-Means indices are not comparable across conditions.

Cluster type	**miRNA only**	**miRNA + clinical**
Basal-Like	MIR-18A/18A* (1.46)	MIR-18A/18A* (1.43)
	MIR-577/577 (1.41)	MIR-505/505 (1.34)
	MIR-505/505 (1.37)	MIR-577/577 (1.33)
	MIR-934/934 (1.32)	MIR-505/505* (1.28)
	MIR-505/505* (1.29)	MIR-224/224* (1.27)
Lum. A (MIR-197 signature)	MIR-197/197 (1.05)	MIR-197/197 (1.12)
	MIR-186/186* (0.99)	MIR-324/5P (1.09)
	MIR-324/5P (0.99)	MIR-186/186* (1.07)
	MIR-146B/5P (0.97)	MIR-146B/5P (1.04)
	MIR-24-1/24 (0.96)	MIR-590/5P (1.02)
Lum. A (MIR-199 signature)	MIR-199B/5P (0.89)	MIR-151/3P (0.81)
	MIR-151/3P (0.79)	MIR-374C/374C (0.78)
	MIR-145/145 (0.76)	MIR-199B/5P (0.77)
	MIR-374C/374C (0.76)	MIR-3607/3P (0.75)
	MIR-3130-1/5P (0.75)	MIR-3130-1/5P (0.74)
Lum. B / HER2 mixed	MIR-33B/33B* (0.88)	MIR-33B/33B* (0.88)
	MIR-3677/3P (0.85)	MIR-3677/3P (0.86)
	MIR-142/5P (0.81)	MIR-589/5P (0.80)
	MIR-326/326 (0.80)	MIR-142/5P (0.79)
	MIR-589/5P (0.80)	MIR-3607/3P (0.79)

Several observations emerge from comparing the signatures across conditions.

The Basal-Like signature is highly stable across conditions. The Basal-Like cluster produces the strongest feature signal in both conditions, with MIR-18A/18A* as the top-ranked feature in both ($z = 1.46$ and $z = 1.43$, respectively). The top-5 features are essentially the same set: MIR-18A/18A*, MIR-505/505, and MIR-577/577 appear in both rankings with nearly identical z-scores. This stability is a key positive result: it confirms that the miRNA signature driving the Basal-Like cluster is robust to the addition of clinical covariates, and that the slight purity drop observed under miRNA + clinical reflects a small shift in which samples are geometrically attracted to the Basal centroid, not a change in the centroid's miRNA character.

One Luminal A cluster is consistently defined by a miR-197-centred signature. The Luminal A cluster with higher z-scores is dominated by MIR-197/197, MIR-186/186*, MIR-324/5P, and MIR-146B/5P in both conditions, with MIR-197/197 ranking first in both. The strong correspondence between conditions for this cluster suggests it captures a biologically robust Luminal A subgroup whose miRNA identity is not perturbed by the addition of clinical features.

The two Luminal A clusters share no features within each condition. Within each condition, the two Luminal A-dominant clusters have completely non-overlapping top-20 feature lists. In the miRNA-only condition, one is characterised by the MIR-197/MIR-186 group (z-scores ≈ 1.0) while the other is defined by MIR-199B, MIR-151, and MIR-145 (z-scores ≈ 0.8). This complete divergence in feature signatures, consistent across both conditions, reinforces the interpretation that the two Luminal A clusters represent genuinely distinct biological subgroups rather than an arbitrary partition of the same population.

The Luminal B / HER2-Enriched mixed cluster shows the weakest and most stable signal. Consistent with its mixed composition and lowest overall purity, the Luminal B / HER2-Enriched cluster exhibits the lowest maximum z-scores of all four clusters in both conditions (< 0.90). Notably, its top features (MIR-33B/33B* and MIR-3677/3P) rank first and second in both conditions with virtually identical z-scores. This stability, paradoxically, reflects the fact that the cluster is primarily defined by what Luminal B and HER2-Enriched samples share in miRNA space, not by any strong subtype-specific signal.

Taken together, the feature analysis confirms that the miRNA-only condition is superior both quantitatively (higher external metrics) and interpretively (no negative-coefficient clusters, cleaner Basal purity). More importantly, the miRNA signatures identified as cluster-defining are largely preserved when clinical features are added, lending confidence to their biological relevance. The comparison of these signatures with the discriminative features from the supervised pipeline is the subject of the following section.

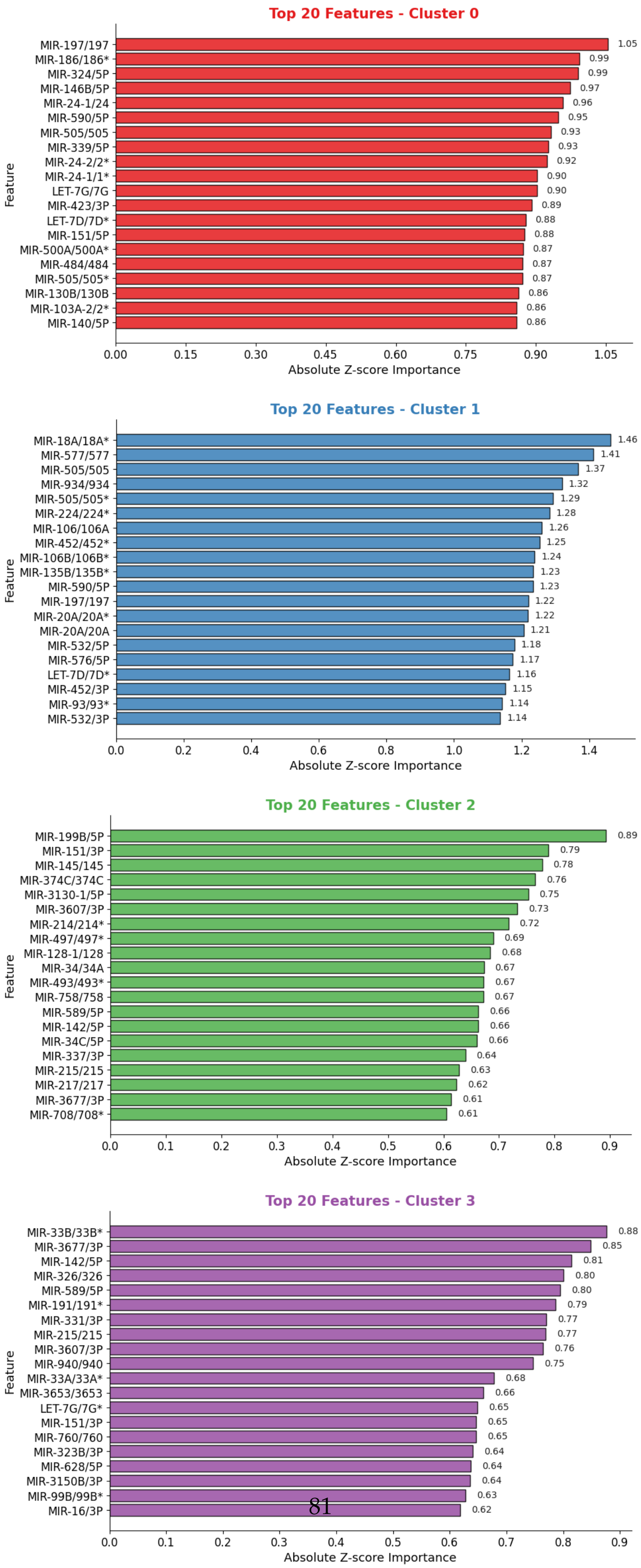


Figure 3.15: Top-20 absolute centroid z-score feature rankings for each K-Means cluster

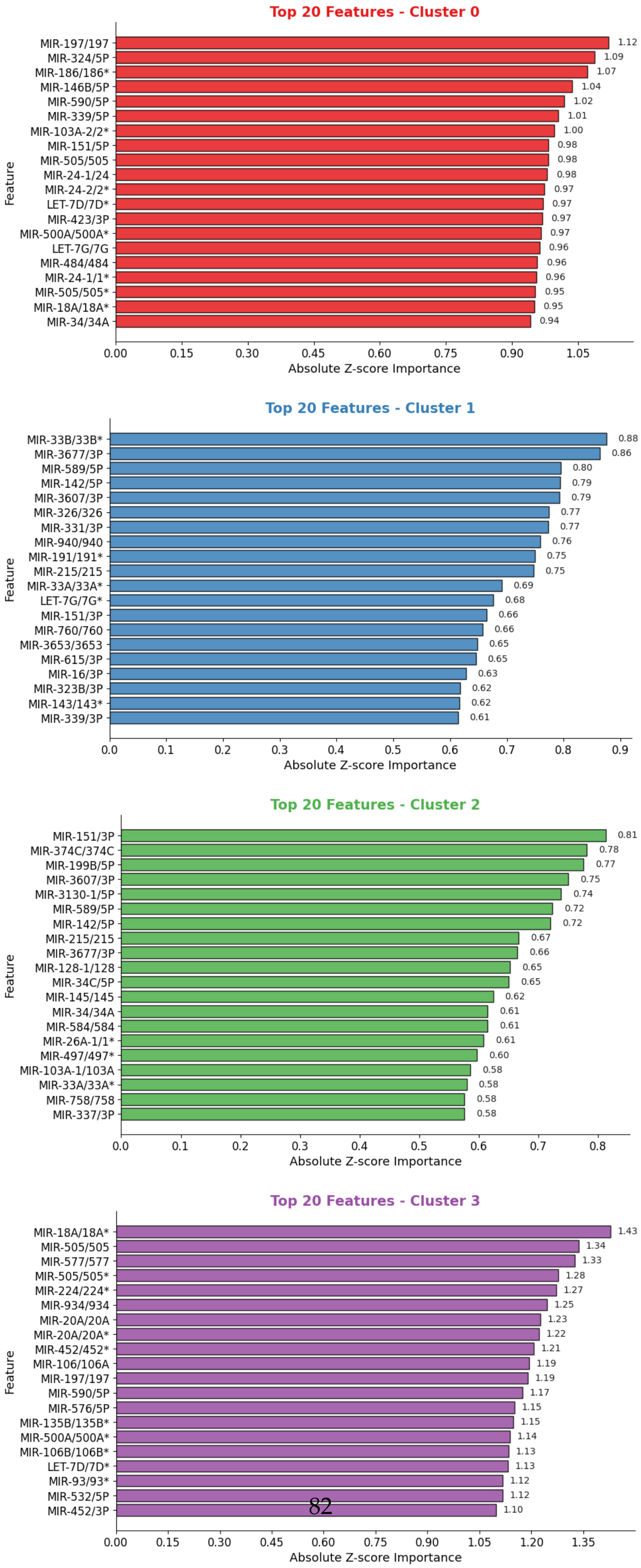


Figure 3.16: Top-20 absolute centroid z-score feature rankings for each K-Means clus-

3.5 Cross-Paradigm Validation: Converging on Subtype-Defining miRNAs

The two analytical paradigms developed in this thesis - supervised classification via XGBoost and ANOVA, and unsupervised clustering via K-Means - were designed to be complementary rather than redundant. The supervised approach asks which miRNAs are discriminative when a model is explicitly trained to separate subtypes; the unsupervised approach asks which miRNAs define the geometry of the data when no labels are consulted. A feature that appears in both answers is therefore supported by two independent, fundamentally different lines of evidence: it is statistically and informationally relevant to a trained classifier, and it is also a structural anchor of the data's intrinsic organisation. This section formalises that cross-paradigm intersection and draws the biological conclusions that follow from it.

For each subtype, the doubly-validated panel is defined as the set of miRNAs that appear in both the panel of doubly-validated features from the superised analysis (composed by the intersection of the ANOVA top-20 ranking and the XGBoost top-20 gain ranking) and the K-Means top-20 centroid z-score ranking for the cluster dominated by that subtype (miRNA-only condition). Tables 3.17 and 3.18 summarise these intersections for the miRNA-only and miRNA+clinical conditions, respectively.

Table 3.17: Cross-paradigm miRNA intersection for each PAM50 subtype. Features listed appear in the top-20 rankings of both XGBoost gain importance and K-Means centroid z-score under the miRNA-only condition. The corresponding unsupervised cluster and its purity are indicated for reference.

Subtype	Unsupervised cluster	Intersection size	Doubly-validated miRNAs
Basal-Like	C1 (72.3% purity)	5/20	MIR-577/577, MIR-135B/135B*, MIR-934/934, MIR-505/505, MIR-18A/18A*
HER2-Enriched	C3 (41.3%, mixed)	≤2/20	MIR-142/5P, MIR-3150B/3P
Luminal A	C0 (66.2%) + C2 (63.6%)	≤2/20 each	MIR-326/326 (C0 only)
Luminal B	C3 (41.3%, mixed)	1/20	MIR-3677/3P

3.5.1 Subtype-by-Subtype Interpretation

The intersection results paint a coherent and biologically grounded picture of the information content of miRNA expression data with respect to PAM50 subtype structure. Rather than producing a uniform result across subtypes, the cross-paradigm analysis reveals a clear gradient: one subtype is robustly characterised, one is partially so, and two remain elusive. This gradient is not a failure of the methodology, it is itself a finding.

Table 3.18: Cross-paradigm miRNA intersection for each PAM50 subtype. Features listed appear in the top-20 rankings of both XGBoost gain importance and K-Means centroid z-score under the miRNA + clinical condition. The corresponding unsupervised cluster and its purity are indicated for reference.

Subtype	Unsupervised cluster	Intersection size	Doubly-validated miRNAs
Basal-Like	C3 (68.0% purity)	7/20	MIR-577/577, MIR-135B/135B*, MIR-934/934, MIR-505/505, MIR-18A/18A*, MIR-20A/20A* MIR-505/505*
HER2-Enriched	C1 (27.7%, mixed)	≤0/20	-
Luminal A	C0 (69.4%) + C2 (64.2%)	≤1/20	MIR-145 / 145 (C2 only)
Luminal B	C1 (41.5%, mixed)	1/20	MIR-3677/3P

Basal-Like: a robust, compact, and convergent miRNA panel. Basal-Like is the only subtype for which both analytical paradigms converge onto a substantial and overlapping feature set. Seven miRNAs, **(MIR-577/577, MIR-135B/135B*, MIR-934/934, MIR-505/505, MIR-18A/18A*, MIR-20A/20A* and MIR-505/505*)**, appear simultaneously in the top-20 XGBoost gain rankings and the top-20 K-Means centroid z-scores for the Basal-dominant cluster under the miRNA-only condition. This is a strong result for several compounding reasons. First, the supervised classifier achieved perfect classification for Basal-Like (F1 = 1.00), and the ANOVA F-scores for its top features (386 for MIR-934, 358 for MIR-577, 292 for MIR-135B) are an order of magnitude above those of any other subtype's top features. Second, the unsupervised Basal-Like cluster is the most geometrically coherent of the four (72.3% purity under miRNA-only, the highest across both conditions), and its centroid z-scores are the highest in the entire analysis (MIR-18A/18A* at $z = 1.46$). Third, the five-miRNA core panel is stable and further enriched under the clinical condition: adding clinical covariates expands the doubly-validated intersection to seven features, incorporating MIR-20A/20A* and MIR-505/505* alongside the original five, without displacing any of them. Taken together, these converging lines of evidence support the conclusion that the doubly-validated miRNAs constitute a compact, robust, and biologically credible panel for Basal-Like subtype identification. This is consistent with the well-established clinical distinctiveness of Basal-Like (triple-negative) tumours, which exhibit a markedly different molecular programme from hormone-receptor-positive subtypes and have been consistently separable in expression-based studies since the earliest molecular classifications of breast cancer [48].

Luminal A: a partially distinctive subtype with intrinsic heterogeneity. Luminal A occupies an intermediate position. The supervised approach identifies a distributed miRNA signature - the gain is spread across ~20 features, with no single dominant marker (top feature contributes only ~5% of gain) - and classification performance is moderate (F1 =

0.81). The unsupervised analysis adds a structural observation that the supervised approach cannot: Luminal A is not a geometrically cohesive entity in miRNA expression space. It consistently fragments into two distinct clusters, each dominated by a completely non-overlapping set of miRNAs. One subgroup is characterised by MIR-197/197, MIR-186/186*, and MIR-324/5P; the other by MIR-199B/5P, MIR-151/3P, and MIR-145/145. The cross-paradigm intersections are accordingly small (≤2 features per cluster) and condition-dependent: under the miRNA-only condition, MIR-326/326 appears as the sole doubly-validated feature for C0, while under the clinical condition MIR-145/145 emerges for C2. The fact that different features are validated across conditions is itself informative, reflecting the instability of any single Luminal A panel rather than a methodological artefact.

This divergence between the supervised and unsupervised feature sets for Luminal A should not be interpreted as a contradiction. The supervised model learns boundaries that maximally separate Luminal A from other subtypes in a multivariate discriminative setting; the unsupervised model identifies the directions of greatest internal variation within the cohort. The fact that these two objectives yield different miRNA rankings for Luminal A is itself informative: it suggests that Luminal A is internally heterogeneous, containing at least two biologically distinct subpopulations whose within-group variation is comparable to their between-group signal against other subtypes. This is consistent with growing evidence in the literature that the Luminal A category, as defined by PAM50, conflates molecularly distinct entities [48]. The practical consequence for biomarker discovery is that a single panel of Luminal A markers is likely to be insufficient; any clinically deployed panel would need to account for this heterogeneity by either targeting the shared boundary features identified by the supervised approach, or by embracing the two-subgroup structure revealed by clustering and defining separate panels for each.

HER2-Enriched: a diffuse signature with no compact discriminative panel. HER2-Enriched is the most analytically challenging subtype, and the cross-paradigm results confirm this systematically. In the supervised analysis, classification performance was the weakest of the four subtypes (F1 = 0.50, with low recall of 0.33), the gain profile is the most distributed of all classes (top feature MIR-30/30A contributes only ~15% of total gain), and the ANOVA F-scores for its top features barely exceed 50, confirming the absence of a statistically dominant, individually discriminative marker. In the unsupervised analysis, HER2-Enriched samples are never captured by a dedicated cluster: they are consistently co-assigned with Luminal B samples in a mixed cluster of approximately 41% purity (28.6% HER2-Enriched composition under miRNA-only; 27.7% under miRNA + clinical). This near-identical composition across conditions confirms that the inseparability of HER2-Enriched and Luminal B is an intrinsic property of the miRNA expression space rather than a consequence of any particular feature set. The cross-paradigm intersection reflects this directly: under the miRNA-only condition it is at most 2 features (MIR-142/5P, MIR-3150B/3P), and under the clinical condition it collapses entirely to zero. The progressive deterioration of the intersection with richer feature sets, rather than improving it, constitutes the strongest possible confirmation that no compact miRNA-based panel exists for this subtype. The conclusion is unambiguous: **miRNA expression alone is insufficient to define a robust discriminative panel for HER2-Enriched tumours.** The supervised results confirm that the most informative supplementary signal for this subtype is `HER2_STATUS`, a clinical covariate that encodes protein-level receptor amplification information that miRNA expression does not capture. Any practical diagnostic panel for HER2-Enriched classification would therefore require the integration of molecular receptor status data alongside miRNA profiling.

Luminal B: no compact panel, boundary inseparability confirmed. Luminal B presents the most challenging scenario of all four subtypes. Supervised classification performance was the lowest (F1 = 0.50 under miRNA-only), improved moderately with the addition of clinical features (F1 = 0.64), and the gain distribution is the most diffuse across all subtypes (top 20 features account for only ~50% of total gain). In the unsupervised analysis, Luminal B shares its cluster with HER2-Enriched samples, contributing 41.3% of cluster composition under the miRNA-only condition and 41.5% under the clinical condition. Despite this analytical difficulty, the cross-paradigm intersection yields one consistent result: MIR-3677/3P is doubly-validated under *both* experimental conditions, appearing in the supervised top-20 for Luminal B and in the unsupervised mixed cluster in each case. This cross-condition stability makes it the single most reliable miRNA anchor for Luminal B, though one feature alone is insufficient to constitute a discriminative panel. The picture is therefore consistent across both paradigms: Luminal B lacks a geometrically compact and statistically concentrated miRNA identity, and its boundary with Luminal A on one side and HER2-Enriched on the other is diffuse in miRNA expression space. As with HER2-Enriched, the supervised results suggest that clinical features (particularly `ER_STATUS`) carry discriminative information that miRNA expression alone does not provide. Unlike HER2-Enriched, however, even the addition of clinical features does not fully resolve the boundary; the improvement in F1-score from 0.50 to 0.64 is meaningful but leaves substantial room for further misclassification.

3.5.2 Summary of Conclusions

The cross-paradigm analysis yields a coherent and internally consistent set of conclusions that address the central question of this thesis: *which miRNAs carry subtype-discriminative information, and what do the two analytical paradigms together tell us about the biological structure of the PAM50 classification?*

The results can be summarised along three axes.

In terms of discriminative miRNA panels: A compact, doubly-validated panel exists for Basal-Like only, comprising MIR-577/577, MIR-135B/135B*, MIR-934/934, MIR-505/505, MIR-18A/18A*, MIR-20A/20A* and MIR-505/505*. For Luminal A, a panel of candidate features can be identified from the supervised analysis, but the unsupervised results caution that this panel describes a boundary rather than a coherent subgroup, and that a two-panel approach accounting for internal heterogeneity may be more appropriate. For HER2-Enriched and Luminal B, no compact miRNA-only panel is supported by either paradigm; their discrimination requires integration of clinical receptor status data.

In terms of the data's intrinsic structure: The unsupervised analysis reveals that the miRNA expression space of this cohort contains one compact and isolated subgroup (Basal-Like), one heterogeneous but partially structured region (Luminal A), and one diffuse region where HER2-Enriched and Luminal B co-occupy indistinct geometric neighbourhoods. This structure is stable across experimental conditions and is not an artefact of any particular algorithmic choice.

In terms of the value of the unsupervised approach: The K-Means analysis adds genuine scientific value beyond what the supervised analysis alone could establish. It confirms the Basal-Like panel through an entirely independent lens; it reveals the internal fragmentation of Luminal A that the classifier treats as a single class; and it places the difficulty of HER2-Enriched and Luminal B classification in a geometric context, showing that these subtypes do not occupy separable regions of miRNA space regardless of the algorithm used. Together, the two paradigms provide a more complete and more trustworthy characterisation of the miRNA landscape of breast cancer subtypes than either could achieve individually.